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| United States Patent | 12402154 |
| Kind Code | B2 |
| Date of Patent | August 26, 2025 |
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## Signaling for inter-base station interference estimation


### Abstract

Methods, systems, and devices for wireless communications are described. A user equipment (UE) may receive a message indicating one or more communication patterns from a base station. The one or more communication patterns may indicate to the UE to adjust one or more parameters for transmitting an uplink communication during a set of time resources, a set of beams, a set of ports, or any combination thereof. During the resources indicated in the one or more communication patterns, the base station may estimate a cross link interference channel between the base station and a second base station. The one or more communication patterns may indicate a hopping pattern for each subband of a bandwidth part. The base station may configure the UE with a timing advance based on the message.



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**Appl. No.:** **18/427282**

**Filed:** **January 30, 2024**


### Prior Publication Data

| Document Identifier | Publication Date |
|---|---|
| US 20240172263 A1 | May. 23, 2024 |

### Related U.S. Application Data

continuation parent-doc US 17567723 20220103 US 11917668 child-doc US 18427282

## Publication Classification

**Int. Cl.:** **H04W4/00** (20180101); **H04W72/541** (20230101)

**U.S. Cl.:**

CPC **H04W72/541** (20230101);

## Field of Classification Search

**CPC:** H04W (72/541); H04W (56/0045); H04W (24/08); H04W (52/146); H04L (5/0012); H04B (7/0695)

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## Background/Summary

CROSS REFERENCE (1) The present application for patent is a continuation of U.S. patent application Ser. No. 17/567,723 by IBRAHIM et al., entitled “SIGNALING FOR INTER-BASE STATION INTERFERENCE ESTIMATION,” filed Jan. 3, 2022, assigned to the assignee hereof, and expressly incorporated by reference herein.

FIELD OF TECHNOLOGY
(1) The following relates to wireless communication, including signaling for inter-base station interference estimation.
BACKGROUND
(2) Wireless communications systems are widely deployed to provide various types of communication content such as voice, video, packet data, messaging, broadcast, and so on. These systems may be capable of supporting communication with multiple users by sharing the available system resources (e.g., time, frequency, and power). Examples of such multiple-access systems include fourth generation (4G) systems such as Long Term Evolution (LTE) systems, LTE-Advanced (LTE-A) systems, or LTE-A Pro systems, and fifth generation (5G) systems which may be referred to as New Radio (NR) systems. These systems may employ technologies such as code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple access (FDMA), orthogonal FDMA (OFDMA), or discrete Fourier transform spread orthogonal frequency division multiplexing (DFT-S-OFDM). A wireless multiple-access communications system may include one or more base stations or one or more network access nodes, each simultaneously supporting communication for multiple communication devices, which may be otherwise known as user equipment (UE).
SUMMARY
(3) The described techniques relate to improved methods, systems, devices, and apparatuses that support signaling for inter-base station interference estimation. Generally, the described techniques provide for measuring interference for full duplex communications. For example, the techniques enable a base station to estimate a cross-link interference (CLI) channel between a first base station and a second base station. Estimating the CLI channel may enable the first base station to mitigate interference between uplink channels and the CLI channel. For example, a user equipment (UE) may receive a message indicating one or more communication patterns from the first base station. In some cases, the one or more communication patterns may indicate to the UE to adjust a parameter (e.g., wait, use a reduced transmit power, mute, timing advance (TA)) for transmitting an uplink communication during a set of time resources, a set of beams, a set of ports, or any combination thereof. During the resources indicated in the one or more communication patterns, the first base station may estimate the CLI channel between the first base station and the second base station. The first base station may then use the estimated CLI channel to improve the reception of uplink communications during full duplex scenarios. In some cases, the one or more

communication patterns may indicate a hopping pattern. The hopping pattern may indicate each subband of a bandwidth part (BWP) that the base station may use to estimate the CLI channel for at least one time resource. In some cases, the base station may configure the UE with a TA based on the message. In some examples, the base station may use one or more types of resources (e.g., a downlink resource or a flexible resource converted to a downlink resource) for measuring CLI.

(4) A method for wireless communication at a UE is described. The method may include receiving, from a first base station, a message that indicates one or more communication patterns of one or more resources for estimating a CLI channel between uplink communications associated with the first base station and downlink communications associated with a second base station, adjusting a transmission parameter for an uplink communication based on receiving the message from the first base station, and transmitting the uplink communication over the one or more resources based on adjusting the transmission parameter.

(5) An apparatus for wireless communication at a UE is described. The apparatus may include a processor, memory coupled with the processor, and instructions stored in the memory. The instructions may be executable by the processor to cause the apparatus to receive, from a first base station, a message that indicates one or more communication patterns of one or more resources for estimating a CLI channel between uplink communications associated with the first base station and downlink communications associated with a second base station, adjust a transmission parameter for an uplink communication based on receiving the message from the first base station, and transmit the uplink communication over the one or more resources based on adjusting the transmission parameter.

(6) Another apparatus for wireless communication at a UE is described. The apparatus may include means for receiving, from a first base station, a message that indicates one or more communication patterns of one or more resources for estimating a CLI channel between uplink communications associated with the first base station and downlink communications associated with a second base station, means for adjusting a transmission parameter for an uplink communication based on receiving the message from the first base station, and means for transmitting the uplink communication over the one or more resources based on adjusting the transmission parameter.

(7) A non-transitory computer-readable medium storing code for wireless communication at a UE is described. The code may include instructions executable by a processor to receive, from a first base station, a message that indicates one or more communication patterns of one or more resources for estimating a CLI channel between uplink communications associated with the first base station and downlink communications associated with a second base station, adjust a transmission parameter for an uplink communication based on receiving the message from the first base station, and transmit the uplink communication over the one or more resources based on adjusting the transmission parameter.

(8) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from the first base station, a second message to activate a communication pattern of the one or more communication patterns, where adjusting a transmission parameter for the uplink communication may be based on receiving the second message.

(9) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the message includes a radio resource control message and the second message includes a medium access control-control element or downlink control information.

(10) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, adjusting the transmission parameter for the uplink communication may include operations, features, means, or instructions for reducing the transmit power of the uplink communications during at least a portion of the one or more resources.

(11) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, reducing the transmit power of the uplink communications may include

operations, features, means, or instructions for reducing the transmit power of the uplink communications to zero during the portion of the one or more resources.

(12) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying a first subband of a BWP for estimating the CLI channel, a second subband of the BWP for a second uplink communication by the UE, and a third subband of the BWP as a guard band between the first subband and the second subband based on receiving the message, the one or more resources including the BWP, where the one or more communication patterns included in the message indicate the first subband, the second subband, and the third subband, where adjusting a transmission parameter for the uplink communication may be based on the identifying.

(13) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting, over the second subband, the uplink communication to the first base station based on transmitting the message.

(14) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying a hopping pattern of subbands of a BWP for estimating the CLI channel and transmitting the uplink communications based on receiving the message, the hopping pattern spanning a set of multiple slots, where the one or more communication patterns indicate the hopping pattern, where adjusting a transmission parameter for the uplink communication may be based on the identifying.

(15) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the hopping pattern indicates at least one time resource for each subband of the subbands of the BWP for estimating the CLI channel.

(16) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, adjusting the transmission parameter for the uplink communication may include operations, features, means, or instructions for applying a TA for the uplink communication transmitted by the UE during a first time resource of the one or more resources based on receiving the message, the TA based on a propagation delay of a downlink communication transmitted by the second base station.

(17) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the TA may be based on a capability of the UE.

(18) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, adjusting the transmission parameter for the uplink communication may include operations, features, means, or instructions for identifying one or more beams to restrict the UE from using to transmit the uplink communication during the one or more resources, where the message indicates the one or more beams.

(19) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from the first base station, a second message that schedules the UE with the uplink communication using the one or more beams and determining to skip the uplink communication or use a different beam than the one or more beams based on the one or more communication patterns.

(20) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying one or more subbands of frequency resources or one or more slots of time resources or both for estimating the CLI channel, where the message includes an indication of the one or more subbands or the one or more slots.

(21) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the one or more resources include one or more subbands of frequency resources and one or more slots of time resources.

(22) In some examples of the method, apparatuses, and non-transitory computer-readable medium

described herein, the one or more communication patterns indicate to the UE to mute or reduce a transmit power of one or more uplink communications during a set of time resources of the one or more resources.
(23) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the one or more communication patterns indicate a set of beams that the UE may be restricted from using for the uplink communications.
(24) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the one or more communication patterns indicate one or more ports for estimating the CLI channel.
(25) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first base station may be configured to receive the uplink communications over the one or more resources and the second base station may be configured to transmit the downlink communications over the one or more resources.
(26) A method for wireless communication at a first base station is described. The method may include transmitting, to a UE, a message that indicates one or more communication patterns of one or more resources for estimating a CLI channel between the first base station and a second base station, the first base station configured to receive uplink communications over the one or more resources, monitoring the one or more resources for CLI associated with the second base station based on transmitting the message, and processing a first uplink communication from the UE communicated over a resource full duplexed with one or more downlink communications associated with the second base station based on monitoring the one or more resources for the CLI.
(27) An apparatus for wireless communication at a first base station is described. The apparatus may include a processor, memory coupled with the processor, and instructions stored in the memory. The instructions may be executable by the processor to cause the apparatus to transmit, to a UE, a message that indicates one or more communication patterns of one or more resources for estimating a CLI channel between the first base station and a second base station, the first base station configured to receive uplink communications over the one or more resources, monitor the one or more resources for CLI associated with the second base station based on transmitting the message, and process a first uplink communication from the UE communicated over a resource full duplexed with one or more downlink communications associated with the second base station based on monitoring the one or more resources for the CLI.
(28) Another apparatus for wireless communication at a first base station is described. The apparatus may include means for transmitting, to a UE, a message that indicates one or more communication patterns of one or more resources for estimating a CLI channel between the first base station and a second base station, the first base station configured to receive uplink communications over the one or more resources, means for monitoring the one or more resources for CLI associated with the second base station based on transmitting the message, and means for processing a first uplink communication from the UE communicated over a resource full duplexed with one or more downlink communications associated with the second base station based on monitoring the one or more resources for the CLI.
(29) A non-transitory computer-readable medium storing code for wireless communication at a first base station is described. The code may include instructions executable by a processor to transmit, to a UE, a message that indicates one or more communication patterns of one or more resources for estimating a CLI channel between the first base station and a second base station, the first base station configured to receive uplink communications over the one or more resources, monitor the one or more resources for CLI associated with the second base station based on transmitting the message, and process a first uplink communication from the UE communicated over a resource full duplexed with one or more downlink communications associated with the second base station based on monitoring the one or more resources for the CLI.
(30) Some examples of the method, apparatuses, and non-transitory computer-readable medium

described herein may further include operations, features, means, or instructions for transmitting a second message to activate a communication pattern of the one or more communication patterns, where monitoring the one or more resources for the CLI may be based on transmitting the second message.
(31) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the message includes a radio resource control message and the second message includes a medium access control-control element or downlink control information.
(32) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying one or more UEs to mute the uplink communications during at least a portion of the one or more resources, the one or more UEs including the first UE, where transmitting the message may be based on identifying the one or more UEs.
(33) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying one or more UEs to reduce a transmit power of the uplink communications during at least a portion of the one or more resources, the one or more UEs including the first UE, where transmitting the message may be based on identifying the one or more UEs.
(34) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying one or more subbands of frequency resources or one or more slots of time resources or both for estimating the CLI channel, where the message includes an indication of the one or more subbands or the one or more slots.
(35) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the one or more resources include one or more subbands of frequency resources and one or more slots of time resources.
(36) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the one or more communication patterns indicate to the UE to mute or reduce a transmit power of one or more uplink communications during a set of time resources of the one or more resources.
(37) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying a first subband of a BWP for estimating the CLI channel, a second subband of the BWP for an uplink communication by the UE, and a third subband of the BWP as a guard band between the first subband and the second subband, the one or more resources including the BWP, where the one or more communication patterns included in the message indicate the first subband, the second subband, and the third subband.
(38) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for monitoring the one or more resources further includes monitoring the first subband of the BWP for the CLI.
(39) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, over the second subband, the uplink communication from the UE based on transmitting the message.
(40) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying a hopping pattern of subbands of a BWP for estimating the CLI channel and receiving the uplink communications, the hopping pattern spanning a set of multiple slots, where the one or more communication patterns indicate the hopping pattern.
(41) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the hopping pattern indicates at least one time resource for each subband of the subbands of the BWP for estimating the CLI channel.

(42) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying a TA for an uplink communication communicated by the UE during the one or more resources, the TA based on a propagation delay of a downlink communication transmitted by the second base station and transmitting, to the UE, a second message that includes the TA, where the monitoring may be based on transmitting the second message.
(43) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the TA may be based on a capability of the UE.
(44) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying one or more beams to restrict the UE from using to transmit the uplink communications during the one or more resources, where the message indicates the one or more beams.
(45) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the one or more communication patterns indicate to the UE a set of beams that the UE may be restricted from using for the uplink communications.
(46) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying a first quantity of reference signals transmitted by the second base station and identifying a second quantity of ports for estimating the CLI channel based on identifying the first quantity of reference signals, where the one or more communication patterns may be based on identifying the second quantity of ports.
(47) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the second base station may be configured to transmit downlink communications over the one or more resources.
(48) A method for wireless communication at a first base station is described. The method may include transmitting a message indicating a first resource including a first type of resource or a second resource including a second type of resource to use for estimating a CLI channel between the first base station and a second base station, where the first base station is configured to receive uplink communications or transmit downlink communications to a UE over the first type of resource and the first base station is configured to transmit the downlink communications to the UE over the second type of resource, monitoring one or more resources for CLI associated with the second base station based on transmitting the message, and processing a first uplink communication from the UE over a full-duplex resource based on monitoring the one or more resources for the CLI.
(49) An apparatus for wireless communication at a first base station is described. The apparatus may include a processor, memory coupled with the processor, and instructions stored in the memory. The instructions may be executable by the processor to cause the apparatus to transmit a message indicating a first resource including a first type of resource or a second resource including a second type of resource to use for estimating a CLI channel between the first base station and a second base station, where the first base station is configured to receive uplink communications or transmit downlink communications to a UE over the first type of resource and the first base station is configured to transmit the downlink communications to the UE over the second type of resource, monitor one or more resources for CLI associated with the second base station based on transmitting the message, and process a first uplink communication from the UE over a full-duplex resource based on monitoring the one or more resources for the CLI.
(50) Another apparatus for wireless communication at a first base station is described. The apparatus may include means for transmitting a message indicating a first resource including a first type of resource or a second resource including a second type of resource to use for estimating a CLI channel between the first base station and a second base station, where the first base station is configured to receive uplink communications or transmit downlink communications to a UE over

the first type of resource and the first base station is configured to transmit the downlink communications to the UE over the second type of resource, means for monitoring one or more resources for CLI associated with the second base station based on transmitting the message, and means for processing a first uplink communication from the UE over a full-duplex resource based on monitoring the one or more resources for the CLI.
(51) A non-transitory computer-readable medium storing code for wireless communication at a first base station is described. The code may include instructions executable by a processor to transmit a message indicating a first resource including a first type of resource or a second resource including a second type of resource to use for estimating a CLI channel between the first base station and a second base station, where the first base station is configured to receive uplink communications or transmit downlink communications to a UE over the first type of resource and the first base station is configured to transmit the downlink communications to the UE over the second type of resource, monitor one or more resources for CLI associated with the second base station based on transmitting the message, and process a first uplink communication from the UE over a full-duplex resource based on monitoring the one or more resources for the CLI.
(52) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting, to the UE, a second message to convert the first type of resource to the second type of resource, where monitoring the one or more resources for the CLI may be based on transmitting the message.
(53) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the second message includes a slot format indicator configured to convert the first resource including the first type of resource to be the second type of resource.
(54) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first type of resource includes a flexible symbol configured to use with the uplink communications or the downlink communications and the second type of resource includes a downlink symbol.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIGS. **1** and **2** illustrate examples of a wireless communications system that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure.
(2) FIG. **3** illustrates an example of a process flow that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure.
(3) FIGS. **4**A and **4**B illustrate examples of configurations that support signaling for inter-base station interference estimation in accordance with aspects of the present disclosure.
(4) FIG. **5** illustrates an example of a wireless communications system timeline that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure.
(5) FIGS. **6** and **7** show block diagrams of devices that support signaling for inter-base station interference estimation in accordance with aspects of the present disclosure.
(6) FIG. **8** shows a block diagram of a communications manager that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure.
(7) FIG. **9** shows a diagram of a system including a device that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure.
(8) FIGS. **10** and **11** show block diagrams of devices that support signaling for inter-base station interference estimation in accordance with aspects of the present disclosure.
(9) FIG. **12** shows a block diagram of a communications manager that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure.

(10) FIG. **13** shows a diagram of a system including a device that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure.
(11) FIGS. **14** through **18** show flowcharts illustrating methods that support signaling for inter-base station interference estimation in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

(12) A base station may support estimating a cross-link interference (CLI) channel to perform operations to mitigate CLI. In some cases, a first base station may receive uplink communication from a first user equipment (UE) over a first set of communication resources. A second base station may transmit downlink communication to a second UE over the same set of communication resources (e.g., using the same time resources and frequency resources). Due to a direction of transmission and a signal strength of the downlink communication, the downlink communication from the second base station may interfere with the uplink transmission from the first UE in a full duplex scenario. In such cases, because of the interference, the first base station may not be able to successfully decode the uplink communication. This interference may be an example of a CLI channel between the first base station and the second base station. For the first base station to perform CLI mitigating procedures, the first base station may estimate the CLI channel (e.g., interference from the second base station on the channel). As the CLI mitigating procedure depends on the CLI estimate, obtaining an accurate CLI estimate is desirable. However, the first base station may receive uplink communications on the channel during a process to estimate the CLI channel, which may degrade the quality (accuracy) of the CLI estimate. While the description of CLI was discussed with reference to a second base station, a first UE, and a second UE, a CLI channel may occur during any signaling interference that a base station may use to receive uplink communications (e.g., from one or more UEs).

(13) The techniques described herein provide procedures for estimating interference for full duplex communications between different nodes in a communication network. The techniques enable a first base station to estimate a CLI channel between the first base station and a second base station that uses the same resources that are used for uplink communications. For example, a UE may receive a message indicating one or more communication patterns from the first base station. In some cases, the one or more communication patterns may indicate to the UE to adjust a parameter (e.g., wait, use a reduced transmit power, mute, timing advance [TA]) for transmitting an uplink communication during a set of time resources, a set of beams, a set of ports, or any combination thereof. During the resources indicated in the one or more communication patterns, the first base station may estimate the CLI channel between the first base station and the second base station. The first base station may then use the estimated CLI channel to improve the reception of uplink communications during full duplex scenarios. In some cases, the one or more communication patterns may indicate a hopping pattern. The hopping pattern may indicate each subband of a BWP that the base station may use to measure CLI for at least one time resource. In some cases, the base station may configure the UE with a TA based on the message. In some examples, the base station may use one or more types of resources (e.g., a downlink resource or a flexible resource converted to a downlink resource) for measuring CLI.

(14) Aspects of the disclosure are initially described in the context of wireless communications systems. Aspects are then described with respect to a process flow, resource configurations, and a wireless communications system timeline. Aspects of the disclosure are further illustrated by and described with reference to apparatus diagrams, system diagrams, and flowcharts that relate to signaling for inter-base station interference estimation.

(15) FIG. **1** illustrates an example of a wireless communications system **100** that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure. The wireless communications system **100** may include one or more base stations **105**, one or more UEs **115**, and a core network **130**. In some examples, the wireless communications system **100** may be a Long Term Evolution (LTE) network, an LTE-Advanced (LTE-A) network, an LTE-A Pro

network, or a New Radio (NR) network. In some examples, the wireless communications system **100** may support enhanced broadband communications, ultra-reliable communications, low latency communications, communications with low-cost and low-complexity devices, or any combination thereof.

(16) The base stations **105** may be dispersed throughout a geographic area to form the wireless communications system **100** and may be devices in different forms or having different capabilities. The base stations **105** and the UEs **115** may wirelessly communicate via one or more communication links **125**. Each base station **105** may provide a coverage area **110** over which the UEs **115** and the base station **105** may establish one or more communication links **125**. The coverage area **110** may be an example of a geographic area over which a base station **105** and a UE **115** may support the communication of signals according to one or more radio access technologies.

(17) The UEs **115** may be dispersed throughout a coverage area **110** of the wireless communications system **100**, and each UE **115** may be stationary, or mobile, or both at different times. The UEs **115** may be devices in different forms or having different capabilities. Some example UEs **115** are illustrated in FIG. **1**. The UEs **115** described herein may be able to communicate with various types of devices, such as other UEs **115**, the base stations **105**, or network equipment (e.g., core network nodes, relay devices, integrated access and backhaul (IAB) nodes, or other network equipment), as shown in FIG. **1**.

(18) In some examples, one or more components of the wireless communications system **100** may operate as or be referred to as a network node. As used herein, a network node may refer to any UE **115**, base station **105**, entity of a core network **130**, apparatus, device, or computing system configured to perform any techniques described herein. For example, a network node may be a UE **115**. As another example, a network node may be a base station **105**. As another example, a first network node may be configured to communicate with a second network node or a third network node. In one aspect of this example, the first network node may be a UE **115**, the second network node may be a base station **105**, and the third network node may be a UE **115**. In another aspect of this example, the first network node may be a UE **115**, the second network node may be a base station **105**, and the third network node may be a base station **105**. In yet other aspects of this example, the first, second, and third network nodes may be different. Similarly, reference to a UE **115**, a base station **105**, an apparatus, a device, or a computing system may include disclosure of the UE **115**, base station **105**, apparatus, device, or computing system being a network node. For example, disclosure that a UE **115** is configured to receive information from a base station **105** also discloses that a first network node is configured to receive information from a second network node. In this example, consistent with this disclosure, the first network node may refer to a first UE **115**, a first base station **105**, a first apparatus, a first device, or a first computing system configured to receive the information; and the second network node may refer to a second UE **115**, a second base station **105**, a second apparatus, a second device, or a second computing system

(19) The base stations **105** may communicate with the core network **130**, or with one another, or both. For example, the base stations **105** may interface with the core network **130** through one or more backhaul links **120** (e.g., via an S1, N2, N3, or other interface). The base stations **105** may communicate with one another over the backhaul links **120** (e.g., via an X2, Xn, or other interface) either directly (e.g., directly between base stations **105**), or indirectly (e.g., via core network **130**), or both. In some examples, the backhaul links **120** may be or include one or more wireless links.

(20) One or more of the base stations **105** described herein may include or may be referred to by a person having ordinary skill in the art as a base transceiver station, a radio base station, an access point, a radio transceiver, a NodeB, an eNodeB (eNB), a next-generation NodeB or a giga-NodeB (either of which may be referred to as a gNB), a Home NodeB, a Home eNodeB, or other suitable terminology.

(21) A UE **115** may include or may be referred to as a mobile device, a wireless device, a remote device, a handheld device, or a subscriber device, or some other suitable terminology, where the

“device” may also be referred to as a unit, a station, a terminal, or a client, among other examples. A UE **115** may also include or may be referred to as a personal electronic device such as a cellular phone, a personal digital assistant (PDA), a tablet computer, a laptop computer, or a personal computer. In some examples, a UE **115** may include or be referred to as a wireless local loop (WLL) station, an Internet of Things (IoT) device, an Internet of Everything (IoE) device, or a machine type communications (MTC) device, among other examples, which may be implemented in various objects such as appliances, or vehicles, meters, among other examples.

(22) The UEs **115** described herein may be able to communicate with various types of devices, such as other UEs **115** that may sometimes act as relays as well as the base stations **105** and the network equipment including macro eNBs or gNBs, small cell eNBs or gNBs, or relay base stations, among other examples, as shown in FIG. **1**.

(23) The UEs **115** and the base stations **105** may wirelessly communicate with one another via one or more communication links **125** over one or more carriers. The term “carrier” may refer to a set of radio frequency spectrum resources having a defined physical layer structure for supporting the communication links **125**. For example, a carrier used for a communication link **125** may include a portion of a radio frequency spectrum band (e.g., a bandwidth part (BWP)) that is operated according to one or more physical layer channels for a given radio access technology (e.g., LTE, LTE-A, LTE-A Pro, NR). Each physical layer channel may carry acquisition signaling (e.g., synchronization signals, system information), control signaling that coordinates operation for the carrier, user data, or other signaling. The wireless communications system **100** may support communication with a UE **115** using carrier aggregation or multi-carrier operation. A UE **115** may be configured with multiple downlink component carriers and one or more uplink component carriers according to a carrier aggregation configuration. Carrier aggregation may be used with both frequency division duplexing (FDD) and time division duplexing (TDD) component carriers.

(24) In some examples (e.g., in a carrier aggregation configuration), a carrier may also have acquisition signaling or control signaling that coordinates operations for other carriers. A carrier may be associated with a frequency channel (e.g., an evolved universal mobile telecommunication system terrestrial radio access (E-UTRA) absolute radio frequency channel number (EARFCN)) and may be positioned according to a channel raster for discovery by the UEs **115**. A carrier may be operated in a standalone mode where initial acquisition and connection may be conducted by the UEs **115** via the carrier, or the carrier may be operated in a non-standalone mode where a connection is anchored using a different carrier (e.g., of the same or a different radio access technology).

(25) The communication links **125** shown in the wireless communications system **100** may include uplink transmissions from a UE **115** to a base station **105**, or downlink transmissions from a base station **105** to a UE **115**. Carriers may carry downlink or uplink communications (e.g., in an FDD mode) or may be configured to carry downlink and uplink communications (e.g., in a TDD mode).

(26) A carrier may be associated with a particular bandwidth of the radio frequency spectrum, and in some examples the carrier bandwidth may be referred to as a “system bandwidth” of the carrier or the wireless communications system **100**. For example, the carrier bandwidth may be one of a number of determined bandwidths for carriers of a particular radio access technology (e.g., 1.4, 3, 5, 10, 15, 20, 40, or 80 megahertz (MHz)). Devices of the wireless communications system **100** (e.g., the base stations **105**, the UEs **115**, or both) may have hardware configurations that support communications over a particular carrier bandwidth or may be configurable to support communications over one of a set of carrier bandwidths. In some examples, the wireless communications system **100** may include base stations **105** or UEs **115** that support simultaneous communications via carriers associated with multiple carrier bandwidths. In some examples, each served UE **115** may be configured for operating over portions (e.g., a sub-band, a BWP) or all of a carrier bandwidth.

(27) Signal waveforms transmitted over a carrier may be made up of multiple subcarriers (e.g.,

using multi-carrier modulation (MCM) techniques such as orthogonal frequency division multiplexing (OFDM) or discrete Fourier transform spread OFDM (DFT-S-OFDM)). In a system employing MCM techniques, a resource element may consist of one symbol period (e.g., a duration of one modulation symbol) and one subcarrier, where the symbol period and subcarrier spacing are inversely related. The number of bits carried by each resource element may depend on the modulation scheme (e.g., the order of the modulation scheme, the coding rate of the modulation scheme, or both). Thus, the more resource elements that a UE **115** receives and the higher the order of the modulation scheme, the higher the data rate may be for the UE **115**. A wireless communications resource may refer to a combination of a radio frequency spectrum resource, a time resource, and a spatial resource (e.g., spatial layers or beams), and the use of multiple spatial layers may further increase the data rate or data integrity for communications with a UE **115**.
(28) One or more numerologies for a carrier may be supported, where a numerology may include a subcarrier spacing (Δf) and a cyclic prefix. A carrier may be divided into one or more BWPs having the same or different numerologies. In some examples, a UE **115** may be configured with multiple BWPs. In some examples, a single BWP for a carrier may be active at a given time and communications for the UE **115** may be restricted to one or more active BWPs.
(29) The time intervals for the base stations **105** or the UEs **115** may be expressed in multiples of a basic time unit which may, for example, refer to a sampling period of T.sub.s=1/(Δf.sub.max.Math.N.sub.f) seconds, where Δf.sub.max may represent the maximum supported subcarrier spacing, and N f may represent the maximum supported discrete Fourier transform (DFT) size. Time intervals of a communications resource may be organized according to radio frames each having a specified duration (e.g., 10 milliseconds (ms)). Each radio frame may be identified by a system frame number (SFN) (e.g., ranging from 0 to 1023).
(30) Each frame may include multiple consecutively numbered subframes or slots, and each subframe or slot may have the same duration. In some examples, a frame may be divided (e.g., in the time domain) into subframes, and each subframe may be further divided into a number of slots. Alternatively, each frame may include a variable number of slots, and the number of slots may depend on subcarrier spacing. Each slot may include a number of symbol periods (e.g., depending on the length of the cyclic prefix prepended to each symbol period). In some wireless communications systems **100**, a slot may further be divided into multiple mini-slots containing one or more symbols. Excluding the cyclic prefix, each symbol period may contain one or more (e.g., N.sub.f) sampling periods. The duration of a symbol period may depend on the subcarrier spacing or frequency band of operation.
(31) A subframe, a slot, a mini-slot, or a symbol may be the smallest scheduling unit (e.g., in the time domain) of the wireless communications system **100** and may be referred to as a transmission time interval (TTI). In some examples, the TTI duration (e.g., the number of symbol periods in a TTI) may be variable. Additionally, or alternatively, the smallest scheduling unit of the wireless communications system **100** may be dynamically selected (e.g., in bursts of shortened TTIs (sTTIs)).
(32) Physical channels may be multiplexed on a carrier according to various techniques. A physical control channel and a physical data channel may be multiplexed on a downlink carrier, for example, using one or more of time division multiplexing (TDM) techniques, frequency division multiplexing (FDM) techniques, or hybrid TDM-FDM techniques. A control region (e.g., a control resource set (CORESET)) for a physical control channel may be defined by a number of symbol periods and may extend across the system bandwidth or a subset of the system bandwidth of the carrier. One or more control regions (e.g., CORESETs) may be configured for a set of the UEs **115**. For example, one or more of the UEs **115** may monitor or search control regions for control information according to one or more search space sets, and each search space set may include one or multiple control channel candidates in one or more aggregation levels arranged in a cascaded manner. An aggregation level for a control channel candidate may refer to a number of control

channel resources (e.g., control channel elements (CCEs)) associated with encoded information for a control information format having a given payload size. Search space sets may include common search space sets configured for sending control information to multiple UEs **115** and UE-specific search space sets for sending control information to a specific UE **115**.

(33) Each base station **105** may provide communication coverage via one or more cells, for example a macro cell, a small cell, a hot spot, or other types of cells, or any combination thereof. The term "cell" may refer to a logical communication entity used for communication with a base station **105** (e.g., over a carrier) and may be associated with an identifier for distinguishing neighboring cells (e.g., a physical cell identifier (PCID), a virtual cell identifier (VCID), or others). In some examples, a cell may also refer to a geographic coverage area **110** or a portion of a geographic coverage area **110** (e.g., a sector) over which the logical communication entity operates. Such cells may range from smaller areas (e.g., a structure, a subset of structure) to larger areas depending on various factors such as the capabilities of the base station **105**. For example, a cell may be or include a building, a subset of a building, or exterior spaces between or overlapping with geographic coverage areas **110**, among other examples.

(34) A macro cell generally covers a relatively large geographic area (e.g., several kilometers in radius) and may allow unrestricted access by the UEs **115** with service subscriptions with the network provider supporting the macro cell. A small cell may be associated with a lower-powered base station **105**, as compared with a macro cell, and a small cell may operate in the same or different (e.g., licensed, unlicensed) frequency bands as macro cells. Small cells may provide unrestricted access to the UEs **115** with service subscriptions with the network provider or may provide restricted access to the UEs **115** having an association with the small cell (e.g., the UEs **115** in a closed subscriber group (CSG), the UEs **115** associated with users in a home or office). A base station **105** may support one or multiple cells and may also support communications over the one or more cells using one or multiple component carriers.

(35) In some examples, a carrier may support multiple cells, and different cells may be configured according to different protocol types (e.g., MTC, narrowband IoT (NB-IoT), enhanced mobile broadband (eMBB)) that may provide access for different types of devices.

(36) In some examples, a base station **105** may be movable and therefore provide communication coverage for a moving geographic coverage area **110**. In some examples, different geographic coverage areas **110** associated with different technologies may overlap, but the different geographic coverage areas **110** may be supported by the same base station **105**. In other examples, the overlapping geographic coverage areas **110** associated with different technologies may be supported by different base stations **105**. The wireless communications system **100** may include, for example, a heterogeneous network in which different types of the base stations **105** provide coverage for various geographic coverage areas **110** using the same or different radio access technologies.

(37) The wireless communications system **100** may support synchronous or asynchronous operation. For synchronous operation, the base stations **105** may have similar frame timings, and transmissions from different base stations **105** may be approximately aligned in time. For asynchronous operation, the base stations **105** may have different frame timings, and transmissions from different base stations **105** may, in some examples, not be aligned in time. The techniques described herein may be used for either synchronous or asynchronous operations.

(38) Some UEs **115**, such as MTC or IoT devices, may be low cost or low complexity devices and may provide for automated communication between machines (e.g., via Machine-to-Machine (M2M) communication). M2M communication or MTC may refer to data communication technologies that allow devices to communicate with one another or a base station **105** without human intervention. In some examples, M2M communication or MTC may include communications from devices that integrate sensors or meters to measure or capture information and relay such information to a central server or application program that makes use of the

information or presents the information to humans interacting with the application program. Some UEs **115** may be designed to collect information or enable automated behavior of machines or other devices. Examples of applications for MTC devices include smart metering, inventory monitoring, water level monitoring, equipment monitoring, healthcare monitoring, wildlife monitoring, weather and geological event monitoring, fleet management and tracking, remote security sensing, physical access control, and transaction-based business charging.

(39) Some UEs **115** may be configured to employ operating modes that reduce power consumption, such as half-duplex communications (e.g., a mode that supports one-way communication via transmission or reception, but not transmission and reception simultaneously). In some examples, half-duplex communications may be performed at a reduced peak rate. Other power conservation techniques for the UEs **115** include entering a power saving deep sleep mode when not engaging in active communications, operating over a limited bandwidth (e.g., according to narrowband communications), or a combination of these techniques. For example, some UEs **115** may be configured for operation using a narrowband protocol type that is associated with a defined portion or range (e.g., set of subcarriers or resource blocks (RBs)) within a carrier, within a guard-band of a carrier, or outside of a carrier.

(40) The wireless communications system **100** may be configured to support ultra-reliable communications or low-latency communications, or various combinations thereof. For example, the wireless communications system **100** may be configured to support ultra-reliable low-latency communications (URLLC). The UEs **115** may be designed to support ultra-reliable, low-latency, or critical functions. Ultra-reliable communications may include private communication or group communication and may be supported by one or more services such as push-to-talk, video, or data. Support for ultra-reliable, low-latency functions may include prioritization of services, and such services may be used for public safety or general commercial applications. The terms ultra-reliable, low-latency, and ultra-reliable low-latency may be used interchangeably herein.

(41) In some examples, a UE **115** may also be able to communicate directly with other UEs **115** over a device-to-device (D2D) communication link **135** (e.g., using a peer-to-peer (P2P) or D2D protocol). One or more UEs **115** utilizing D2D communications may be within the geographic coverage area **110** of a base station **105**. Other UEs **115** in such a group may be outside the geographic coverage area **110** of a base station **105** or be otherwise unable to receive transmissions from a base station **105**. In some examples, groups of the UEs **115** communicating via D2D communications may utilize a one-to-many (1:M) system in which each UE **115** transmits to every other UE **115** in the group. In some examples, a base station **105** facilitates the scheduling of resources for D2D communications. In other cases, D2D communications are carried out between the UEs **115** without the involvement of a base station **105**.

(42) In some systems, the D2D communication link **135** may be an example of a communication channel, such as a sidelink communication channel, between vehicles (e.g., UEs **115**). In some examples, vehicles may communicate using vehicle-to-everything (V2X) communications, vehicle-to-vehicle (V2V) communications, or some combination of these. A vehicle may signal information related to traffic conditions, signal scheduling, weather, safety, emergencies, or any other information relevant to a V2X system. In some examples, vehicles in a V2X system may communicate with roadside infrastructure, such as roadside units, or with the network via one or more network nodes (e.g., base stations **105**) using vehicle-to-network (V2N) communications, or with both.

(43) The core network **130** may provide user authentication, access authorization, tracking, Internet Protocol (IP) connectivity, and other access, routing, or mobility functions. The core network **130** may be an evolved packet core (EPC) or 5G core (5GC), which may include at least one control plane entity that manages access and mobility (e.g., a mobility management entity (MME), an access and mobility management function (AMF)) and at least one user plane entity that routes packets or interconnects to external networks (e.g., a serving gateway (S-GW), a Packet Data

Network (PDN) gateway (P-GW), or a user plane function (UPF)). The control plane entity may manage non-access stratum (NAS) functions such as mobility, authentication, and bearer management for the UEs **115** served by the base stations **105** associated with the core network **130**. User IP packets may be transferred through the user plane entity, which may provide IP address allocation as well as other functions. The user plane entity may be connected to IP services **150** for one or more network operators. The IP services **150** may include access to the Internet, Intranet(s), an IP Multimedia Subsystem (IMS), or a Packet-Switched Streaming Service.
(44) Some of the network devices, such as a base station **105**, may include subcomponents such as an access network entity **140**, which may be an example of an access node controller (ANC). Each access network entity **140** may communicate with the UEs **115** through one or more other access network transmission entities **145**, which may be referred to as radio heads, smart radio heads, or transmission/reception points (TRPs). Each access network transmission entity **145** may include one or more antenna panels. In some configurations, various functions of each access network entity **140** or base station **105** may be distributed across various network devices (e.g., radio heads and ANCs) or consolidated into a single network device (e.g., a base station **105**).
(45) The wireless communications system **100** may operate using one or more frequency bands, typically in the range of 300 megahertz (MHz) to 300 gigahertz (GHz). Generally, the region from 300 MHz to 3 GHz is known as the ultra-high frequency (UHF) region or decimeter band because the wavelengths range from approximately one decimeter to one meter in length. The UHF waves may be blocked or redirected by buildings and environmental features, but the waves may penetrate structures sufficiently for a macro cell to provide service to the UEs **115** located indoors. The transmission of UHF waves may be associated with smaller antennas and shorter ranges (e.g., less than 100 kilometers) compared to transmission using the smaller frequencies and longer waves of the high frequency (HF) or very high frequency (VHF) portion of the spectrum below 300 MHz.
(46) The wireless communications system **100** may also operate in a super high frequency (SHF) region using frequency bands from 3 GHz to 30 GHz, also known as the centimeter band, or in an extremely high frequency (EHF) region of the spectrum (e.g., from 30 GHz to 300 GHz), also known as the millimeter band. In some examples, the wireless communications system **100** may support millimeter wave (mmW) communications between the UEs **115** and the base stations **105**, and EHF antennas of the respective devices may be smaller and more closely spaced than UHF antennas. In some examples, this may facilitate use of antenna arrays within a device. The propagation of EHF transmissions, however, may be subject to even greater atmospheric attenuation and shorter range than SHF or UHF transmissions. The techniques disclosed herein may be employed across transmissions that use one or more different frequency regions, and designated use of bands across these frequency regions may differ by country or regulating body.
(47) The wireless communications system **100** may utilize both licensed and unlicensed radio frequency spectrum bands. For example, the wireless communications system **100** may employ License Assisted Access (LAA), LTE-Unlicensed (LTE-U) radio access technology, or NR technology in an unlicensed band such as the 5 GHz industrial, scientific, and medical (ISM) band. When operating in unlicensed radio frequency spectrum bands, devices such as the base stations **105** and the UEs **115** may employ carrier sensing for collision detection and avoidance. In some examples, operations in unlicensed bands may be based on a carrier aggregation configuration in conjunction with component carriers operating in a licensed band (e.g., LAA). Operations in unlicensed spectrum may include downlink transmissions, uplink transmissions, P2P transmissions, or D2D transmissions, among other examples.
(48) A base station **105** or a UE **115** may be equipped with multiple antennas, which may be used to employ techniques such as transmit diversity, receive diversity, multiple-input multiple-output (MIMO) communications, or beamforming. The antennas of a base station **105** or a UE **115** may be located within one or more antenna arrays or antenna panels, which may support MIMO operations or transmit or receive beamforming. For example, one or more base station antennas or antenna

arrays may be co-located at an antenna assembly, such as an antenna tower. In some examples, antennas or antenna arrays associated with a base station **105** may be located in diverse geographic locations. A base station **105** may have an antenna array with a number of rows and columns of antenna ports that the base station **105** may use to support beamforming of communications with a UE **115**. Likewise, a UE **115** may have one or more antenna arrays that may support various MIMO or beamforming operations. Additionally, or alternatively, an antenna panel may support radio frequency beamforming for a signal transmitted via an antenna port.

(49) The base stations **105** or the UEs **115** may use MIMO communications to exploit multipath signal propagation and increase the spectral efficiency by transmitting or receiving multiple signals via different spatial layers. Such techniques may be referred to as spatial multiplexing. The multiple signals may, for example, be transmitted by the transmitting device via different antennas or different combinations of antennas. Likewise, the multiple signals may be received by the receiving device via different antennas or different combinations of antennas. Each of the multiple signals may be referred to as a separate spatial stream and may carry bits associated with the same data stream (e.g., the same codeword) or different data streams (e.g., different codewords). Different spatial layers may be associated with different antenna ports used for channel measurement and reporting. MIMO techniques include single-user MIMO (SU-MIMO), where multiple spatial layers are transmitted to the same receiving device, and multiple-user MIMO (MU-MIMO), where multiple spatial layers are transmitted to multiple devices.

(50) Beamforming, which may also be referred to as spatial filtering, directional transmission, or directional reception, is a signal processing technique that may be used at a transmitting device or a receiving device (e.g., a base station **105**, a UE **115**) to shape or steer an antenna beam (e.g., a transmit beam, a receive beam) along a spatial path between the transmitting device and the receiving device. Beamforming may be achieved by combining the signals communicated via antenna elements of an antenna array such that some signals propagating at particular orientations with respect to an antenna array experience constructive interference while others experience destructive interference. The adjustment of signals communicated via the antenna elements may include a transmitting device or a receiving device applying amplitude offsets, phase offsets, or both to signals carried via the antenna elements associated with the device. The adjustments associated with each of the antenna elements may be defined by a beamforming weight set associated with a particular orientation (e.g., with respect to the antenna array of the transmitting device or receiving device, or with respect to some other orientation).

(51) A base station **105** or a UE **115** may use beam sweeping techniques as part of beam forming operations. For example, a base station **105** may use multiple antennas or antenna arrays (e.g., antenna panels) to conduct beamforming operations for directional communications with a UE **115**. Some signals (e.g., synchronization signals, reference signals, beam selection signals, or other control signals) may be transmitted by a base station **105** multiple times in different directions. For example, the base station **105** may transmit a signal according to different beamforming weight sets associated with different directions of transmission. Transmissions in different beam directions may be used to identify (e.g., by a transmitting device, such as a base station **105**, or by a receiving device, such as a UE **115**) a beam direction for later transmission or reception by the base station **105**.

(52) Some signals, such as data signals associated with a particular receiving device, may be transmitted by a base station **105** in a single beam direction (e.g., a direction associated with the receiving device, such as a UE **115**). In some examples, the beam direction associated with transmissions along a single beam direction may be determined based on a signal that was transmitted in one or more beam directions. For example, a UE **115** may receive one or more of the signals transmitted by the base station **105** in different directions and may report to the base station **105** an indication of the signal that the UE **115** received with a highest signal quality or an otherwise acceptable signal quality.

(53) In some examples, transmissions by a device (e.g., by a base station **105** or a UE **115**) may be performed using multiple beam directions, and the device may use a combination of digital precoding or radio frequency beamforming to generate a combined beam for transmission (e.g., from a base station **105** to a UE **115**). The UE **115** may report feedback that indicates precoding weights for one or more beam directions, and the feedback may correspond to a configured number of beams across a system bandwidth or one or more sub-bands. The base station **105** may transmit a reference signal (e.g., a cell-specific reference signal (CRS), a channel state information reference signal (CSI-RS)), which may be precoded or unprecoded. The UE **115** may provide feedback for beam selection, which may be a precoding matrix indicator (PMI) or codebook-based feedback (e.g., a multi-panel type codebook, a linear combination type codebook, a port selection type codebook). Although these techniques are described with reference to signals transmitted in one or more directions by a base station **105**, a UE **115** may employ similar techniques for transmitting signals multiple times in different directions (e.g., for identifying a beam direction for subsequent transmission or reception by the UE **115**) or for transmitting a signal in a single direction (e.g., for transmitting data to a receiving device).
(54) A receiving device (e.g., a UE **115**) may try multiple receive configurations (e.g., directional listening) when receiving various signals from the base station **105**, such as synchronization signals, reference signals, beam selection signals, or other control signals. For example, a receiving device may try multiple receive directions by receiving via different antenna subarrays, by processing received signals according to different antenna subarrays, by receiving according to different receive beamforming weight sets (e.g., different directional listening weight sets) applied to signals received at multiple antenna elements of an antenna array, or by processing received signals according to different receive beamforming weight sets applied to signals received at multiple antenna elements of an antenna array, any of which may be referred to as “listening” according to different receive configurations or receive directions. In some examples, a receiving device may use a single receive configuration to receive along a single beam direction (e.g., when receiving a data signal). The single receive configuration may be aligned in a beam direction determined based on listening according to different receive configuration directions (e.g., a beam direction determined to have a highest signal strength, highest signal-to-noise ratio (SNR), or otherwise acceptable signal quality based on listening according to multiple beam directions).
(55) The wireless communications system **100** may be a packet-based network that operates according to a layered protocol stack. In the user plane, communications at the bearer or Packet Data Convergence Protocol (PDCP) layer may be IP-based. A Radio Link Control (RLC) layer may perform packet segmentation and reassembly to communicate over logical channels. A Medium Access Control (MAC) layer may perform priority handling and multiplexing of logical channels into transport channels. The MAC layer may also use error detection techniques, error correction techniques, or both to support retransmissions at the MAC layer to improve link efficiency. In the control plane, the Radio Resource Control (RRC) protocol layer may provide establishment, configuration, and maintenance of an RRC connection between a UE **115** and a base station **105** or a core network **130** supporting radio bearers for user plane data. At the physical layer, transport channels may be mapped to physical channels.
(56) The UEs **115** and the base stations **105** may support retransmissions of data to increase the likelihood that data is received successfully. Hybrid automatic repeat request (HARQ) feedback is one technique for increasing the likelihood that data is received correctly over a communication link **125**. HARQ may include a combination of error detection (e.g., using a cyclic redundancy check (CRC)), forward error correction (FEC), and retransmission (e.g., automatic repeat request (ARQ)). HARQ may improve throughput at the MAC layer in poor radio conditions (e.g., low signal-to-noise conditions). In some examples, a device may support same-slot HARQ feedback, where the device may provide HARQ feedback in a specific slot for data received in a previous symbol in the slot. In other cases, the device may provide HARQ feedback in a subsequent slot, or

according to some other time interval.
(57) In some examples, the wireless communications system **100** may support a type of full duplex communication for different base stations. Full duplex communication may include a communication device transmitting and receiving on a same time resource, a same frequency resource, or any combination thereof. If the communication device transmits and receives on the same time resource and the same frequency resource, the full duplex communication may be considered as in-band full duplex (IBFD). In IBFD, the downlink and uplink communications may share a same IBFD time and frequency resource such that the downlink and uplink communications are either partially or fully overlapped across the resources. If the communication device transmits and receives on the same time resource but on different frequency resources the full duplex communication may be considered as a sub-band flexible duplex (FDD) or sub-band full duplex (SBFD). In sub-band FDD the downlink resource may be separated from the uplink resource in the frequency domain by a guard band (GB).
(58) In some cases, signal interference may come from inter-base station or inter-UE communication in various forms of full duplex operation. For example, a first base station may be able to configure a set of time resources for uplink or downlink communications and a second base station may be able to configure the same set of time resources for uplink or downlink communications, independently of the first base station. In situations where the first base station configures the set of resources for uplink communications and the second base station configures the set of resources for downlink communications, the first base station may experience CLI during the set of resources. In another example, a first base station **105** and a second base station **105** may be in full duplex operation and a first UE **115** and a second UE **115** may be in half duplex operation. The first base station **105** may transmit a downlink communication to the first UE **115** while receiving uplink communication from the second UE **115**, causing self-interference. During a same resource, the second base station may transmit a downlink communication to either the first UE **115** or the second UE **115** and cause inter-base station interference. In one example, the first base station **105**, the second base station **105**, the first UE **115**, and the second UE **115** may be in full duplex operation. The first UE **115** may transmit and receive wireless communication, causing self-interference. During the same resource, the second UE **115** may receive a downlink communication from the second base station **105** and cause inter-UE interference. In one example, the first base station **105** and the second base station **105** may be in half duplex operation and the first UE **115** may be in full duplex operation. The first UE **115** may transmit an uplink communication to the first base station **105** and receive a downlink communication from the second base station **105** during a same resource, causing self-interference. The various forms on interference may be examples of CLI.
(59) In some examples, CLI management (e.g., similar to dynamic TDD scenarios) may be desirable. There may be one or more CLI mitigating procedures that may apply, or be effective, in different scenarios. For example, for intra operator interference (e.g., co-channel), during inter-base station interference, the one or more CLI mitigating procedures may include spatial separation, BWP partitioning for frequency division multiplexing (FDM), and interference cancellation (e.g., beamforming nulling and optional digital interference cancellation). During inter-UE interference, the one or more CLI mitigating procedures may include CLI aided scheduling. For inter operator interference (e.g., adjacent channel), during inter-base station interference, the one or more CLI mitigating procedures may include max uplink and downlink BWP separation and spatial separation of co-located antennas. During inter-UE interference, the one or more CLI mitigating procedures may include max uplink and downlink BWP separation.
(60) In some examples of inter-base station CLI mitigation, a first base station **105** and a second base station **105** may coordinate. For example, in co-channel interference (e.g., dynamic TDD), a coverage area **110** (e.g., a neighboring cell or sector) may coordinate to reduce interference by restricting one or more beams, power back-off, dynamic zoning, slot conversion, or any

combination thereof. In some cases, the first base station **105** and the second base station **105** may coordinate for receive beamforming (nulling) or interference cancellation that may require an inter-base station channel to project into null space.

(61) In some cases, to perform the various CLI mitigating procedures or techniques, a victim base station **105** may estimate a CLI channel (e.g., from an aggressor base station **105**). In some cases, a process that incorporates measuring for CLI may include performing receiver or transmitter nulling (e.g., beamforming nulling and digital interference cancellation) based on knowing the channel (e.g., per-tone), estimating dominant direction (beam) from the aggressor base station **105** in the channel, and finding a combiner configuration based on inter-base station CLI. In some cases, the CLI estimate may include an inter-base station channel and aggressor base station samples for the interference cancelation. As the CLI mitigating procedures depend on the CLI measurement or estimate, obtaining an accurate CLI estimate is desirable. However, the victim base station **105** may receive uplink communications on the channel during the CLI measurement or estimation, which may degrade the quality (accuracy) of the CLI measurement or estimation. To increase the accuracy of the CLI measurement or estimate at the victim base station **105**, the victim base station **105** may send, to one or more connected UEs **115**, a communication pattern to request the one or more connected UEs **115** to adjust a transmission parameter for uplink communications during at least a portion of time for measuring or estimating CLI. In some examples, the communication pattern may be an uplink muting indication. The uplink muting indication may prevent the one or more connected UEs **115** from transmitting uplink communications within resources used for CLI measurement or estimation, as the uplink communications may interfere with the CLI channel measurement. In some cases, simultaneous CLI channel measurement and reception of uplink communications may be possible in SBFD slots, provided enough isolation in frequency to mitigate the impact of timing misalignment.

(62) In some examples, in an SBFD scenario, the timing misalignment for CLI channel measurement may cause an inaccurate CLI measurement. For example, the time that the victim base station **105** receives the uplink communication and the time that the victim base station **105** measures the CLI channel for interference from the aggressor base station **105** (e.g., a downlink communication from the aggressor base station **105**) may be different. Because the one or more connected UEs **115** have different propagation delays than the aggressor base station **105** a cyclic prefix (CP) of the uplink communication and a second CP of the downlink communication may not overlap in time. Thus, an orthogonality between the communications may be misaligned and the quality of an estimation of CLI on the channel may be reduced. In some examples, the different propagation delays may be due to each UE **115** having a different TA while the aggressor base station **105** has no TA and each device being located at a different distance from the victim base station **105**, among other examples.

(63) In some cases, an uplink cancellation indication (ULCI) may indicate to the one or more UEs **115** to cancel a transmission in one or more resources (e.g., time and frequency). For example, the victim base station **105** may send downlink control information (DCI) 2_4 to a UE **115** to indicate to the UE **115** to cancel PUSCH (SRS) transmissions in one or more time resource, one or more frequency resource, or any combination thereof. However, this indication (e.g., L1 signaling) may introduce L1 overhead, which may cause an inefficient use of communication resources and reduce user experience.

(64) In some examples, the described techniques may enable a base station **105** to estimate or measure CLI over resources that may also be used for uplink communications while mitigating interference from other communications. For example, a UE **115** may receive a message indicating one or more communication patterns from the base station **105**. In some cases, the one or more communication patterns may indicate to the UE **115** to adjust a parameter (e.g., wait, use a reduced transmit power, mute, TA) for transmitting an uplink communication during a set of time resources, a set of beams, a set of ports, or any combination thereof. During the resources indicated in the one

or more communication patterns, the base station **105** may estimate or measure the CLI. The base station **105** may then use the CLI to improve the reception of uplink communications during full duplex scenarios. In some cases, the one or more communication patterns may indicate a hopping pattern. The hopping pattern may indicate each subband of a BWP that the base station **105** may use to measure CLI for at least one time resource. In some cases, the base station **105** may configure the UE **115** with a TA based on the message. In some examples, the base station **105** may use one or more types of resources (e.g., a downlink resource or a flexible resource converted to a downlink resource) for measuring CLI.
(65) FIG. **2** illustrates an example of a wireless communications system **200** that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure. In some examples, the wireless communications system **200** may implement aspects of the wireless communications system **100**. The wireless communications system **200** may include a UE **115**-*a*, a base station **105**-*a* and a base station **105**-*b*, which may be examples of a UE **115** and a base station **105** respectively, as described herein with reference to FIG. **1**. In some cases, the base station **105**-*a* may be an example of a victim base station **105** and the base station **105**-*b* may be an example of an aggressor base station **105**, as described herein with reference to FIG. **1**.
(66) In some examples, wireless communications sent by the base station **105**-*b* may interfere with wireless communications sent by the UE **115**-*a* to the base station **105**-*a*. For example, the base station **105**-*a* may receive uplink communication from the UE **115**-*a* over an uplink beam **205**. Using the same time and frequency resources, the base station **105**-*b* may transmit over a downlink beam **210**. When the base station **105**-*b* transmits over the downlink beam **210** in a same direction as the UE **115**-*a* transmits over the uplink beam **205**, then the transmission on the downlink beam **210** may interfere with the transmission on the uplink beam **205** (e.g., CLI). In some cases, the strength of the interference may be inconsistent (e.g., signal strength, direction of transmission). Due to various variables (e.g., environmental effects, transmit power, other interference), a signal strength of the downlink beam **210** may interfere with the uplink beam **205** at differing levels. In some cases, the closer the base station **105**-*b* transmits over the downlink beam **210** in the same direction as the uplink beam **205** the greater the CLI may be. While the description of CLI is discussed with reference to the UE **115**-*a* and the base station **105**-*b*, CLI may occur during any signaling interference on a channel that the base station **105**-*a* may use to receive uplink communications (e.g., from one or more UEs **115**) as described with reference to FIG. **1**.
(67) In some cases, the base station **105**-*a* may measure or estimate the CLI channel between the base station **105**-*a* and the base station **105**-*b* in order to perform CLI mitigating procedures. The base station **105**-*a* may transmit a message **215** to the UE **115**-*a*. In some cases, the message **215** may indicate one or more communication patterns for one or more resources. The one or more communication patterns may include a set of hopping patterns, a set of muting patterns, or any combination thereof. The set of muting patterns may indicate a parameter adjustment (e.g., wait, use a reduced transmit power, mute) for uplink communications. For example, the one or more communication patterns may include a set of muting patterns that indicate time resources (symbols) where the UE **115**-*a* may cancel uplink transmissions (e.g., transmissions over uplink beam **205**). The set of muting patterns may be configured as wide-band or subband-based. In some cases, the muting pattern may include one or more resources due to an uplink timing from UE **115**-*a* being different from a downlink timing from the base station **105**-*b*, as described in more detail with reference to FIG. **5**.
(68) In some examples, the one or more resources may be time resources, frequency resources, or any combination thereof, where the one or more resources may be for measuring CLI on the channel (e.g., inter-base station CLI channel measurement). The one or more resources may be semi-statically (e.g., periodic or semi-persistent) configured and indicated using layer two (L2) or layer three (L3) signaling. For example, the set of muting patterns may be RRC configured. The one or more resources may be periodic, such that the CLI channel measurement may occur with a

certain frequency. In one example, one or more patterns of the set of muting patterns may be activated or deactivated by L2 signaling (e.g., MAC-CE or DCI) for a semi-persistent configuration.

(69) In some cases, the one or more communication patterns indicated by the message **215** may include an indication of a set of uplink beams to restrict. For example, the base station **105**-*a* may determine to restrict the UE **115**-*a* from transmitting over the uplink beam **205**. Because the UE **115**-*a* may transmit over the uplink beam **205** in the same direction as the base station **105**-*b* may transmit over the downlink beam **210**. However, the UE **115**-*a* may transmit over a second uplink beam **220** as the UE **115**-*a* may transmit over the second uplink beam **220** in a different direction from the downlink beam **210**. In some examples, restricting the set of uplink beams may allow for simultaneous uplink receive and CLI channel measurement in a same subband (e.g., for IBFD scenarios), or may allow for isolation in SBFD.

(70) In some cases, the base station **105**-*a* may define a set of restricted uplink beams (e.g., the uplink beam **205**). For example, the one or more communication patterns may indicate the set of restricted uplink beams. The set of restricted uplink beams may indicate to the UE **115**-*a* which uplink beams to refrain from transmitting over during CLI channel measurement occasions. In the case that the UE **115**-*a* is scheduled to transmit uplink information over one or more restricted uplink beams of the set of restricted uplink beams, the UE **115**-*a* may be pre-configured or configured to skip the transmission or to fallback to transmitting over some preconfigured default beam. In one example, to facilitate restricting the set of uplink beams, the base station **105**-*a* may use a legacy spatial relation framework.

(71) In some examples, the one or more communication patterns indicated by the message **215** may include an indication of a set of ports. For example, the CLI channel measurement may be a single-port or multi-port measurement. In some cases, the base station **105**-*b* may transmit various downlink signals (e.g., a synchronization signal block (SSB), a multi-port CSI-RS, a physical data shared channel-demodulation reference signal (PDSCH-DMRS)) over the downlink beam **210**. The quantity of ports indicated in the set of ports may be based on the various downlink signals. In some cases, the base station **105**-*a* may configure different measurement occasions for single-port and multi-port CLI channel measurement.

(72) In some cases, the base station **105**-*a* may select a resource type to use for CLI channel measurement. In one example, the base station **105**-*a* may select a first resource type semi-statically configured as a downlink resource (e.g., in TDD-UL-DL-configCommon). During the first resource type the UE **115**-*a* may refrain from transmitting over the uplink beam **205**. In some examples, the base station **105**-*a* may transmit a configuration message (e.g., an RRC-Reconfiguration message) to change which resources to use for the CLI channel measurement.

(73) In one example, the base station **105**-*a* may select a second resource type (e.g., a flexible resource) to use for the CLI channel measurement. For example, the base station may select the second resource type and transmit control signaling (e.g., a DCI, an SFI) to convert the second resource type to the first resource type (e.g., a downlink resource). The base station **105**-*a* may use the converted resource for CLI channel measurement. In some cases, the second resource type may be configured as a flexible resource in TDD-UL-DL-configCommon, TDD-UL-DL-configDedicated, or any combination thereof. Because the first resource type is used for measuring the CLI over the channel and the UE **115**-*a* may refrain from transmitting uplink communications, the accuracy of the CLI measurement may be improved.

(74) In some examples, the base station **105**-*a* may transmit the control signaling (e.g., a DCI with an SFI) to change the one or more communication patterns. The UE **115**-*a* may buffer a receive signal on the first resource type and the second resource type and resources that intersect with a search space for which the UE **115**-*a* may perform blind decodes for physical downlink control channel (PDCCH).

(75) FIG. **3** illustrates an example of a process flow **300** that supports signaling for inter-base

station interference estimation in accordance with aspects of the present disclosure. The process flow **300** may include a UE **115**-*b*, a base station **105**-*c*, and a base station **105**-*d*, which may be respective examples of UEs **115** and base stations **105** as described with reference to FIGS. **1** and **2**. In some cases, the base station **105**-*d* may be an example of a victim base station **105** and the base station **105**-*c* may be an example of an aggressor base station **105**, as described herein with reference to FIGS. **1** and **2**.

(76) Optionally, at **305**, the base station **105**-*d* may select one or more resources for CLI estimation. In some examples, the selected resources may include one or more subbands of frequency resources and one or more slots of time resources. The selected resources may be resources associated with the base station **105**-*c* and the UE **115**-*b*. For example, the base station **105**-*c* may transmit downlink communications in the direction of the base station **105**-*d* using the selected resources. The UE **115**-*b* may transmit uplink communications to the base station **105**-*d* using the selected resources. Because the downlink communications and the uplink communications are using the selected resources, the downlink communications may interfere with the uplink communications at the base station **105**-*d* (e.g., CLI). The base station **105**-*d* may measure or estimate the CLI channel between the base station **105**-*c* and the base station **105**-*d* to perform CLI mitigating procedures.

(77) At **310**, the base station **105**-*d* may transmit a message to at least the UE **115**-*b*. In some cases, the message may include an RRC message and indicate one or more communication patterns of the selected resources for estimating the CLI channel. The one or more communication patterns may include a set of hopping patterns, a set of muting patterns, or any combination thereof. The set of muting patterns may indicate a parameter adjustment (e.g., wait, use a reduced transmit power, mute) for uplink communications.

(78) In some examples, a hopping pattern of the set of hopping patterns may indicate a pattern of subbands across one or more slots of a BWP for estimating the CLI channel. The pattern may indicate at least one resource (e.g., a time resource) for each subband of the subbands to be used for estimating the CLI channel, such that each subband of the BWP is measured. The indicated resource for estimating the CLI channel may hop from subband to subband, across the one or more slots, until each subband of the BWP has been measured, as described in more detail with reference to FIG. **4**. The hopping pattern may allow for an SBFD full duplex scenario, where a slot may contain subbands for CLI channel estimation and downlink or uplink communication. In some cases, to mitigate potential interference, a guard band may be included between the subband for CLI channel estimation and the subband for downlink or uplink communication.

(79) Optionally, at **315**, the base station **105**-*d* may transmit an activation message. The activation message may include L2 signaling (e.g., MAC-CE or DCI) for activating or deactivating a communication pattern of the one or more communication patterns. For example, the communication pattern may be activated by the base station **105**-*d* before estimating the CLI channel and then deactivated.

(80) Optionally, at **320**, the base station **105**-*d* may transmit a TA message. In some cases, the TA message may comprise a TA for CLI channel estimation. For example, the downlink communications and the uplink communications may be misaligned due to different propagation delays. Because of the misalignment, the CLI channel estimation may be inaccurate. To align the communications the base station **105**-*d* may identify a TA for the uplink communications such that an orthogonality between the communications may be aligned, as described in more detail with reference to FIG. **5**. Thus, the accuracy of the CLI channel estimation may be improved. In some cases, the TA may at least be based on a capability of the UE **115**-*b*.

(81) Optionally, at **325**, the UE **115**-*b* may adjust an uplink communications parameter based on the indicated parameter adjustment for uplink communications using the selected resources. For example, the UE **115**-*b* may reduce the transmit power of the uplink communications during at least a portion of the selected resources. In some cases, the UE **115**-*b* may reduce the transmit

power to an amount greater than zero, to zero, or an amount that is associated with the CLI being below a threshold (e.g., a percent of the original CLI). In one example, the UE **115**-*b* may wait to transmit the uplink communications for an amount of time or skip the uplink communications.

(82) At **330**, the base station **105**-*d* may monitor the selected resources for the CLI associated with the base station **105**-*c*. In some cases, the base station **105**-*d* may identify a BWP for estimating the CLI channel. The base station **105**-*d* may monitor subbands of the BWP for the CLI channel estimation. In one example, the base station **105**-*d* may monitor the subbands based on a hopping pattern of the set of hopping patterns, such that each subband of the BWP is monitored across one or more slots.

(83) At **335**, the base station **105**-*d* may receive the downlink communications from the base station **105**-*c* as part of monitoring the resources. The downlink communications may interfere with the uplink communications at the base station **105**-*d* (e.g., CLI). In some cases, the strength of the interference may be inconsistent (e.g., signal strength, direction of transmission). Due to various variables (e.g., environmental effects, transmit power, other interference), a signal strength of the downlink communications may interfere with the uplink communications at differing levels. In some cases, the closer the base station **105**-*c* transmits the downlink communications in the same direction as the uplink communications the greater the CLI.

(84) Optionally, at **340**, the UE **115**-*b* may transmit the uplink communications to the base station **105**-*d* over the selected resources. These uplink communications may be transmitted using an adjust parameter indicated by the message **310**. In some cases, the UE **115**-*b* may transmit the uplink communications using the selected resources, on designated subbands based on the hopping patterns, using the TA for CLI channel estimation, at a reduced transmit power, or any combination thereof. In some cases, the UE **115**-*b* may refrain from transmitting the uplink communications over one or more beams indicated by the one or more communication patterns. The UE **115**-*b* may determine to skip the uplink communications or use a different beam than the one or more beams.

(85) At **345**, the base station **105**-*d* may estimate or measure the CLI channel between the base station **105**-*d* and the base station **105**-*c*. The CLI between the two base stations may be used in future communications to improve the reliability of decoding and account for the CLI.

(86) At **350**, the base station **105**-*d* may process other uplink communications over resources using the estimated CLI channel. In some cases, the resource may be full duplexed with one or more of the downlink communications associated with the base station **105**-*c*. The base station **105**-*d* may process the uplink communications based on the estimating.

(87) FIGS. **4**A and **4**B illustrate examples of a communication pattern **400**-*a* and a communication pattern **400**-*b* that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure. The communication pattern **400**-*a* and the communication pattern **400**-*b* may be examples of the one or more communication patterns of the wireless communications system **100**, the wireless communications system **200**, and the process flow **300**, as described with reference to FIGS. **1-3**. For example, one or both of the communication pattern **400**-*a* and the communication pattern **400**-*b* may be implemented by one or both of a base station **105** or a UE **115** to support measuring interference for full duplex communications. While the communication pattern **400**-*a* and the communication pattern **400**-*b* are illustrated as examples, there may be many more similar patterns to measure CLI channel interference that are supported by the described techniques herein.

(88) In some examples, a base station **105** may measure a channel for CLI and receive uplink communications. However, the received uplink communications may be misaligned in time with the CLI channel measurement, which may cause inter-channel interference (ICI) and reduce the quality (accuracy) of the CLI channel measurement. In some cases, a base station **105** may configure CLI channel measurements in SBFD slots of a BWP. A UE **115** may transmit the uplink communications over an SBFD slot at a first subband. The base station **105** may measure for CLI over the SBFD slot at a second subband (e.g., a same time occasion) to perform CLI mitigating

procedures based on the measurement. Because the base station **105** configures the uplink communications and the CLI channel measurement on different subbands (e.g., the first subband and the second subband), frequency isolation between the different subbands may mitigate the ICI caused by time misalignment between the uplink communications and the CLI channel measurement. To further isolate the different subbands, the base station **105** may configure guard bands between uplink subbands, downlink subbands, subbands for CLI measurement, or any combination thereof. In some examples, to measure CLI over each subband of the BWP, the base station **105** may configure hopping patterns.

(89) For example, the base station **105** may transmit a message (e.g., the message **215**) that indicates one or more communication patterns for one or more slots and resources of the BWP. In order to receive uplink communications (e.g., the uplink beam **205**) and simultaneously measure for CLI on a channel (e.g., the BWP) the one or more communication patterns may indicate a hopping pattern. For example, the hoping pattern may relate to a BWP **405**-*a* or a BWP **405**-*b*. The BWPs **405**-*a* and **405**-*b* may include multiple slots and resources. In some cases, the multiple slots may include half duplex slots, full duplex (e.g., SBFD) slots, or both. The SBFD full duplex slots may include various subbands for CLI measurement resources, uplink resources, downlink resources, and guard bands (sometimes referred to as GBs).

(90) In the example of FIG. **4**A, the BWP **405**-*a* may include various half duplex and full duplex slots. In some examples, the base station **105** may measure each subband of the BWP **405**-*a* for an accurate CLI measurement. The hopping pattern may indicate different SBFD full duplex slot configurations such that each subband includes a CLI channel measurement occasion for at least one time resource. A slot **410**-*a* and a slot **410**-*b* may be examples of an SBFD full duplex slot. The slots **410**-*a* and **410**-*b* may include resources for CLI channel measurement, downlink communication, and uplink communication. For example, the slot **410**-*a* may include an uplink signal **415**-*a* at a second subband. The slot **410**-*a* may further include a downlink signal **420**-*a* at a first subband and a downlink signal **420**-*b* at a third subband. The downlink signals **420**-*a* and **420**-*b* may include CLI measurement occasions **430**-*a* and **430**-*b* respectively. To increase frequency isolation, and thereby increase the accuracy of the CLI channel estimation, the slot **410**-*a* may include a GB **425**-*a* and a GB **425**-*b* between the uplink signal **415**-*a* and the downlink signals **420**-*a* and **420**-*b*.

(91) The slot **410**-*b* may include the resources for CLI channel measurement, downlink communication, and uplink communication in a different configuration such that each subband of the BWP is measured for CLI. For example, the slot **410**-*b* may include uplink signals **415**-*b* and **415**-*c* at the first and third subbands respectively. The slot **410**-*a* may further include a downlink signal **420**-*c* at the second subband, where the downlink signal **420**-*c* may include a CLI measurement occasion **430**-*c*. To promote frequency isolation, the slot **410**-*b* may include a GB **425**-*c* and a GB **425**-*d* between the uplink signals **415**-*b* and **415**-*c* and the downlink signal **420**-*c*. In some cases, the CLI measurement occasions **430**-*a*, **430**-*b*, and **430**-*c* consist of one or more resources in their respective subband (e.g., two symbols).

(92) In the example of FIG. **4**B, the BWP **405**-*b* may include various half duplex and full duplex slots. In some examples, a half-duplex slot may refer to a slot that is assigned to uplink communications or downlink communications for the base stations or cells that could interfere with one another. In some examples, a full-duplex slot may refer to a slot that is configurable for uplink or downlink by different base stations. Various combinations of slot configurations may cause CLI in a full-duplex slot. In some examples, the base station **105** may measure each subband of the BWP **405**-*b* for an accurate CLI measurement. The hopping pattern may indicate different SBFD full duplex slot configurations such that each subband includes a CLI channel measurement occasion for at least one time resource. A slot **410**-*c*, a slot **410**-*d*, and a slot **410**-*e* may be examples of an SBFD full duplex slot. The slots **410**-*c*, **410**-*d*, and **410**-*e* may include resources for CLI channel measurement, downlink communication, and uplink communication. For example, the

slot **410**-*c* may include a downlink signal **420**-*d* at a first subband, an uplink signal **415**-*d* at a second subband, and a CLI measurement occasion **430**-*d* at a third subband. To promote frequency isolation, and thereby increase the accuracy of the CLI channel measurement, the slot **410**-*c* may include a GB **425**-*e* between the uplink signal **415**-*d* and the CLI measurement occasion **430**-*d*. The slot **410**-*d* may include a downlink signal **420**-*e* at the third subband, an uplink signal **415**-*e* at the second subband, and a CLI measurement occasion **430**-*e* at the first subband. To promote frequency isolation, the slot **410**-*d* may include a GB **425**-*f* between the uplink signal **415**-*e* and the CLI measurement occasion **430**-*e*. The slot **410**-*e* may include a downlink signal **420**-*f* at the third subband, an uplink signal **415**-*f* at the first subband, and a CLI measurement occasion **430**-*f* at the second subband. To promote frequency isolation, the slot **410**-*e* may include a GB **425**-*g* between the uplink signal **415**-*f* and the CLI measurement occasion **430**-*f*. In some examples, the CLI measurement occasions **430**-*d*, **430**-*e*, and **430**-*f* may span their entire slot (e.g., each time resource of their respective slot). In these cases, the CLI measurement occasions **430**-*d*, **430**-*e*, and **430**-*f* may use different receive beams.

(93) FIG. **5** illustrates an example of a timing diagram **500** that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure. The timing diagram **500** includes a first base station **105**, a second base station **105**, and a UE **115**, which may be examples of a base station **105** and a UE **115** respectively, as described herein with reference to FIG. **1**. In some cases, the first base station **105** may be an example of a victim base station **105** and the second base station **105** may be an example of an aggressor base station **105**, as described herein with reference to FIGS. **1** and **2**. The timing diagram **500** includes a first TA configuration **525**-*a* and a second TA configuration **525**-*b*. In some examples, the first TA configuration **525**-*a* may be an example of communication with CLI and the second TA configuration **525**-*b* may be an example of a CLI measuring or estimating procedure.

(94) In some cases, in an SBFD scenario, the timing misalignment for CLI channel measurement or estimation may cause an inaccurate CLI measurement. A time resource that the first base station **105** receives an uplink communication may be different than a time resource that the first base station **105** measures a channel for CLI from the second base station **105** (e.g., a downlink communication from the second base station **105**). For example, the second base station **105** may transmit a downlink message **515**-*a* at a time t.sub.1. The UE **115** may transmit an uplink message **520**-*a* at a time t.sub.2, to the first base station **105**, using a first TA **505**-*a*. The first base station **105** may transmit a downlink message **516**-*a* to the UE **115** at a time t.sub.3, where the first TA **505**-*a* may be an advance offset compared to the downlink message **516**-*a*. By applying a TA to first transmission, both the first transmission and a second transmission may arrive at a receiving device at or near the same time. A first distance from the UE **115** to the first base station **105** may be less than a second distance from the second base station **105** to the first base station **105**. Because the first distance is less than the second distance a propagation delay **510**-*a* may be shorter in time than a propagation delay **510**-*b*. The first base station **105** may receive the uplink message **520**-*a* at a time t.sub.4 and may receive the downlink message **515**-*a* at a time t.sub.5 based on the different propagation delays **510**-*a* and **510**-*b*. In some cases, the first distance may be less than, greater than, or equal to the second distance. Because the UE **115** has a different delay than the second base station **105**, a first CP of the uplink communication and a second CP of the downlink communication may not overlap in time. Thus, an orthogonality between the communications may be misaligned and the quality of an estimation of CLI on the channel may be reduced. In some examples, the different propagation delays may be due to the UE **115** having a different TA while the second base station **105** has no TA and each device being located at a different distance from the first base station **105**, among other examples.

(95) In some examples, the time t.sub.4 may be equal to the time t.sub.3, the time t.sub.1 may be equal to the time t.sub.3 (e.g., if a downlink transmission timing is aligned at the first base station **105** and the second base station **105**), or any combination thereof.

(96) In order to reduce the offset between the uplink communication and the CLI channel measurement timing, the first base station **105** may configure one or more UEs **115** with different TA values for CLI measurement occasions. By reducing the offset, an ICI between the uplink communication and the CLI channel measurement may be reduced. For example, to align the orthogonality between the communications, the first base station **105** may transmit a message (e.g., the message **215**) to the UE **115**. The message may include a second TA **505-*b*** based on the CLI measuring procedure. The UE **115** may use the second TA **505-*b*** for determining at what time to transmit an uplink message **520-*b***. The first base station **105** may calculate the second TA **505-*b*** based on a propagation delay **510-*a*** and a propagation delay **510-*b*** of the UE **115** and the second base station **105** respectively.
(97) Thus, the first base station **105** may calculate the second TA **505-*b*** such that the orthogonality between the communications may be aligned. For example, at time t.sub.6, the second base station **105** may transmit a downlink message **515-*b***. The UE **115** may transmit an uplink message **520-*b*** at a time t.sub.7 based on the second TA **505-*b***. The first base station **105** may transmit a downlink message **516-*b*** to the UE **115** at a time t.sub.8, where the second TA **505-*b*** may be an advance offset compared to the downlink message **516-*b***. After the propagation delay **510-*a*** and the propagation delay **510-*b***, at a time t.sub.9, the first base station **105** may receive the uplink message **520-*b*** and the downlink message **515-*b***. In some cases, the first base station **105** may receive the uplink message **520-*b*** and the downlink message **515-*b*** at different times, but with overlapping time resources during their respective CP. Because the CPs overlap in time, the orthogonality between the messages may be aligned and the quality of the estimation of CLI may be improved.
(98) In some cases, the second TA **505-*b*** may be a zero or negative TA value to reduce the timing offset with the CLI channel measurement. If the second TA **505-*b*** is the zero or negative TA value, consecutive uplink and downlink resource scheduling for the UE **115** may be restricted. For example, the first base station **105** may configure a guard resource between the consecutive downlink resource and the consecutive uplink resource for the UE **115**.
(99) In some cases, the UE **115** may apply the second TA **505-*b***. Thus, a capability of the UE **115** may determine the applicability of the second TA **505-*b*** as a procedure for measuring CLI on the channel. For example, the UE **115** may transmit UE capability information to the first base station **105**. The UE capability information may indicate that the UE **115** may apply the second TA **505-*b***. The first base station **105** may determine whether to transmit the second TA **505-*b*** based on the indication.
(100) FIG. **6** shows a block diagram **600** of a device **605** that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure. The device **605** may be an example of aspects of a UE **115** as described herein. The device **605** may include a receiver **610**, a transmitter **615**, and a communications manager **620**. The device **605** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).
(101) The receiver **610** may provide a means for receiving information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to signaling for inter-base station interference estimation). Information may be passed on to other components of the device **605**. The receiver **610** may utilize a single antenna or a set of multiple antennas.
(102) The transmitter **615** may provide a means for transmitting signals generated by other components of the device **605**. For example, the transmitter **615** may transmit information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to signaling for inter-base station interference estimation). In some examples, the transmitter **615** may be co-located with a receiver **610** in a transceiver module. The transmitter **615** may utilize a single

antenna or a set of multiple antennas.
(103) The communications manager **620**, the receiver **610**, the transmitter **615**, or various combinations thereof or various components thereof may be examples of means for performing various aspects of signaling for inter-base station interference estimation as described herein. For example, the communications manager **620**, the receiver **610**, the transmitter **615**, or various combinations or components thereof may support a method for performing one or more of the functions described herein.
(104) In some examples, the communications manager **620**, the receiver **610**, the transmitter **615**, or various combinations or components thereof may be implemented in hardware (e.g., in communications management circuitry). The hardware may include a processor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other programmable logic device, a discrete gate or transistor logic, discrete hardware components, or any combination thereof configured as or otherwise supporting a means for performing the functions described in the present disclosure. In some examples, a processor and memory coupled with the processor may be configured to perform one or more of the functions described herein (e.g., by executing, by the processor, instructions stored in the memory).
(105) Additionally or alternatively, in some examples, the communications manager **620**, the receiver **610**, the transmitter **615**, or various combinations or components thereof may be implemented in code (e.g., as communications management software or firmware) executed by a processor. If implemented in code executed by a processor, the functions of the communications manager **620**, the receiver **610**, the transmitter **615**, or various combinations or components thereof may be performed by a general-purpose processor, a DSP, a central processing unit (CPU), an ASIC, an FPGA, or any combination of these or other programmable logic devices (e.g., configured as or otherwise supporting a means for performing the functions described in the present disclosure).
(106) In some examples, the communications manager **620** may be configured to perform various operations (e.g., receiving, monitoring, transmitting) using or otherwise in cooperation with the receiver **610**, the transmitter **615**, or both. For example, the communications manager **620** may receive information from the receiver **610**, send information to the transmitter **615**, or be integrated in combination with the receiver **610**, the transmitter **615**, or both to receive information, transmit information, or perform various other operations as described herein.
(107) The communications manager **620** may support wireless communication at a UE in accordance with examples as disclosed herein. For example, the communications manager **620** may be configured as or otherwise support a means for receiving, from a first base station, a message that indicates one or more communication patterns of one or more resources for estimating a CLI channel between uplink communications associated with the first base station and downlink communications associated with a second base station. The communications manager **620** may be configured as or otherwise support a means for adjusting a transmission parameter for an uplink communication based on receiving the message from the first base station. The communications manager **620** may be configured as or otherwise support a means for transmitting the uplink communication over the one or more resources based on adjusting the transmission parameter.
(108) By including or configuring the communications manager **620** in accordance with examples as described herein, the device **605** (e.g., a processor controlling or otherwise coupled to the receiver **610**, the transmitter **615**, the communications manager **620**, or a combination thereof) may support techniques for more efficient utilization of communication resources and more accurate CLI channel estimation.
(109) FIG. **7** shows a block diagram **700** of a device **705** that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure. The device **705** may be an example of aspects of a device **605** or a UE **115** as described herein. The device **705** may include a receiver **710**, a transmitter **715**, and a communications manager **720**. The device **705**

may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).

(110) The receiver **710** may provide a means for receiving information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to signaling for inter-base station interference estimation). Information may be passed on to other components of the device **705**. The receiver **710** may utilize a single antenna or a set of multiple antennas.

(111) The transmitter **715** may provide a means for transmitting signals generated by other components of the device **705**. For example, the transmitter **715** may transmit information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to signaling for inter-base station interference estimation). In some examples, the transmitter **715** may be co-located with a receiver **710** in a transceiver module. The transmitter **715** may utilize a single antenna or a set of multiple antennas.

(112) The device **705**, or various components thereof, may be an example of means for performing various aspects of signaling for inter-base station interference estimation as described herein. For example, the communications manager **720** may include a messaging manager **725**, an interference manager **730**, an uplink manager **735**, or any combination thereof. The communications manager **720** may be an example of aspects of a communications manager **620** as described herein. In some examples, the communications manager **720**, or various components thereof, may be configured to perform various operations (e.g., receiving, monitoring, transmitting) using or otherwise in cooperation with the receiver **710**, the transmitter **715**, or both. For example, the communications manager **720** may receive information from the receiver **710**, send information to the transmitter **715**, or be integrated in combination with the receiver **710**, the transmitter **715**, or both to receive information, transmit information, or perform various other operations as described herein.

(113) The communications manager **720** may support wireless communication at a UE in accordance with examples as disclosed herein. The messaging manager **725** may be configured as or otherwise support a means for receiving, from a first base station, a message for estimating a CLI channel between uplink communications associated with the first base station and downlink communications associated with a second base station. The interference manager **730** may be configured as or otherwise support a means for adjusting a transmission parameter for an uplink communication based on receiving the message from the first base station. The uplink manager **735** may be configured as or otherwise support a means for transmitting the uplink communication over the one or more resources based on adjusting the transmission parameter.

(114) FIG. **8** shows a block diagram **800** of a communications manager **820** that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure. The communications manager **820** may be an example of aspects of a communications manager **620**, a communications manager **720**, or both, as described herein. The communications manager **820**, or various components thereof, may be an example of means for performing various aspects of signaling for inter-base station interference estimation as described herein. For example, the communications manager **820** may include a messaging manager **825**, an interference manager **830**, an uplink manager **835**, a power manager **840**, a frequency manager **845**, a hopping pattern manager **850**, a TA manager **855**, a beam manager **860**, or any combination thereof. Each of these components may communicate, directly or indirectly, with one another (e.g., via one or more buses).

(115) The communications manager **820** may support wireless communication at a UE in accordance with examples as disclosed herein. The messaging manager **825** may be configured as or otherwise support a means for receiving, from a first base station, a message for estimating a CLI channel between uplink communications associated with the first base station and downlink communications associated with a second base station. The interference manager **830** may be

configured as or otherwise support a means for adjusting a transmission parameter for an uplink communication based on receiving the message from the first base station. The uplink manager **835** may be configured as or otherwise support a means for transmitting the uplink communication over the one or more resources based on adjusting the transmission parameter.
(116) In some examples, the messaging manager **825** may be configured as or otherwise support a means for receiving, from the first base station, a second message to activate a communication pattern of the one or more communication patterns, where adjusting a transmission parameter for the uplink communication is based on receiving the second message.
(117) In some examples, the message includes a radio resource control message. In some examples, the second message includes a medium access control-control element or downlink control information.
(118) In some examples, to support adjusting the transmission parameter for the uplink communication, the power manager **840** may be configured as or otherwise support a means for reducing the transmit power of the uplink communications during at least a portion of the one or more resources.
(119) In some examples, to support reducing the transmit power of the uplink communications, the power manager **840** may be configured as or otherwise support a means for reducing the transmit power of the uplink communications to zero during the portion of the one or more resources.
(120) In some examples, the frequency manager **845** may be configured as or otherwise support a means for identifying a first subband of a BWP for estimating the CLI channel, a second subband of the BWP for a second uplink communication by the UE, and a third subband of the BWP as a guard band between the first subband and the second subband based on receiving the message, the one or more resources including the BWP, where the one or more communication patterns included in the message indicate the first subband, the second subband, and the third subband, where adjusting a transmission parameter for the uplink communication is based on the identifying.
(121) In some examples, the frequency manager **845** may be configured as or otherwise support a means for transmitting, over the second subband, the uplink communication to the first base station based on transmitting the message.
(122) In some examples, the hopping pattern manager **850** may be configured as or otherwise support a means for identifying a hopping pattern of subbands of a BWP for estimating the CLI channel and transmitting the uplink communications based on receiving the message, the hopping pattern spanning a set of multiple slots, where the one or more communication patterns indicate the hopping pattern, where adjusting a transmission parameter for the uplink communication is based on the identifying.
(123) In some examples, the hopping pattern indicates at least one time resource for each subband of the subbands of the BWP for estimating the CLI channel.
(124) In some examples, to support adjusting the transmission parameter for the uplink communication, the TA manager **855** may be configured as or otherwise support a means for applying a TA for the uplink communication transmitted by the UE during a first time resource of the one or more resources based on receiving the message, the TA based on a propagation delay of a downlink communication transmitted by the second base station. In some examples, the TA is based on a capability of the UE.
(125) In some examples, to support adjusting the transmission parameter for the uplink communication, the beam manager **860** may be configured as or otherwise support a means for identifying one or more beams to restrict the UE from using to transmit the uplink communication during the one or more resources, where the message indicates the one or more beams.
(126) In some examples, the beam manager **860** may be configured as or otherwise support a means for receiving, from the first base station, a second message that schedules the UE with the uplink communication using the one or more beams. In some examples, the beam manager **860** may be configured as or otherwise support a means for determining to skip the uplink

communication or use a different beam than the one or more beams based on the one or more communication patterns.
(127) In some examples, the frequency manager **845** may be configured as or otherwise support a means for identifying one or more subbands of frequency resources or one or more slots of time resources or both for estimating the CLI channel, where the message includes an indication of the one or more subbands or the one or more slots.
(128) In some examples, the one or more resources include one or more subbands of frequency resources and one or more slots of time resources.
(129) In some examples, the one or more communication patterns indicate to the UE to mute or reduce a transmit power of one or more uplink communications during a set of time resources of the one or more resources. In some examples, the one or more communication patterns indicate a set of beams that the UE is restricted from using for the uplink communications. In some examples, the one or more communication patterns indicate one or more ports for estimating the CLI channel.
(130) In some examples, the first base station is configured to receive the uplink communications over the one or more resources. In some examples, the second base station is configured to transmit the downlink communications over the one or more resources.
(131) FIG. **9** shows a diagram of a system **900** including a device **905** that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure. The device **905** may be an example of or include the components of a device **605**, a device **705**, or a UE **115** as described herein. The device **905** may communicate wirelessly with one or more base stations **105**, UEs **115**, or any combination thereof. The device **905** may include components for bi-directional voice and data communications including components for transmitting and receiving communications, such as a communications manager **920**, an input/output (I/O) controller **910**, a transceiver **915**, an antenna **925**, a memory **930**, code **935**, and a processor **940**. These components may be in electronic communication or otherwise coupled (e.g., operatively, communicatively, functionally, electronically, electrically) via one or more buses (e.g., a bus **945**).
(132) The I/O controller **910** may manage input and output signals for the device **905**. The I/O controller **910** may also manage peripherals not integrated into the device **905**. In some cases, the I/O controller **910** may represent a physical connection or port to an external peripheral. In some cases, the I/O controller **910** may utilize an operating system such as iOS®, ANDROID®, MS-DOS®, MS-WINDOWS®, OS/2®, UNIX®, LINUX®, or another known operating system. Additionally, or alternatively, the I/O controller **910** may represent or interact with a modem, a keyboard, a mouse, a touchscreen, or a similar device. In some cases, the I/O controller **910** may be implemented as part of a processor, such as the processor **940**. In some cases, a user may interact with the device **905** via the I/O controller **910** or via hardware components controlled by the I/O controller **910**.
(133) In some cases, the device **905** may include a single antenna **925**. However, in some other cases, the device **905** may have more than one antenna **925**, which may be capable of concurrently transmitting or receiving multiple wireless transmissions. The transceiver **915** may communicate bi-directionally, via the one or more antennas **925**, wired, or wireless links as described herein. For example, the transceiver **915** may represent a wireless transceiver and may communicate bi-directionally with another wireless transceiver. The transceiver **915** may also include a modem to modulate the packets, to provide the modulated packets to one or more antennas **925** for transmission, and to demodulate packets received from the one or more antennas **925**. The transceiver **915**, or the transceiver **915** and one or more antennas **925**, may be an example of a transmitter **615**, a transmitter **715**, a receiver **610**, a receiver **710**, or any combination thereof or component thereof, as described herein.
(134) The memory **930** may include random access memory (RAM) and read-only memory (ROM). The memory **930** may store computer-readable, computer-executable code **935** including instructions that, when executed by the processor **940**, cause the device **905** to perform various

functions described herein. The code **935** may be stored in a non-transitory computer-readable medium such as system memory or another type of memory. In some cases, the code **935** may not be directly executable by the processor **940** but may cause a computer (e.g., when compiled and executed) to perform functions described herein. In some cases, the memory **930** may contain, among other things, a basic I/O system (BIOS) which may control basic hardware or software operation such as the interaction with peripheral components or devices.

(135) The processor **940** may include an intelligent hardware device (e.g., a general-purpose processor, a DSP, a CPU, a microcontroller, an ASIC, an FPGA, a programmable logic device, a discrete gate or transistor logic component, a discrete hardware component, or any combination thereof). In some cases, the processor **940** may be configured to operate a memory array using a memory controller. In some other cases, a memory controller may be integrated into the processor **940**. The processor **940** may be configured to execute computer-readable instructions stored in a memory (e.g., the memory **930**) to cause the device **905** to perform various functions (e.g., functions or tasks supporting signaling for inter-base station interference estimation). For example, the device **905** or a component of the device **905** may include a processor **940** and memory **930** coupled to the processor **940**, the processor **940** and memory **930** configured to perform various functions described herein.

(136) The communications manager **920** may support wireless communication at a UE in accordance with examples as disclosed herein. For example, the communications manager **920** may be configured as or otherwise support a means for receiving, from a first base station, a message for estimating a CLI channel between uplink communications associated with the first base station and downlink communications associated with a second base station. The communications manager **920** may be configured as or otherwise support a means for adjusting a transmission parameter for an uplink communication based on receiving the message from the first base station. The communications manager **920** may be configured as or otherwise support a means for transmitting the uplink communication over the one or more resources based on adjusting the transmission parameter.

(137) By including or configuring the communications manager **920** in accordance with examples as described herein, the device **905** may support techniques for more efficient utilization of communication resources and improved user experience related to reduced interference.

(138) In some examples, the communications manager **920** may be configured to perform various operations (e.g., receiving, monitoring, transmitting) using or otherwise in cooperation with the transceiver **915**, the one or more antennas **925**, or any combination thereof. Although the communications manager **920** is illustrated as a separate component, in some examples, one or more functions described with reference to the communications manager **920** may be supported by or performed by the processor **940**, the memory **930**, the code **935**, or any combination thereof. For example, the code **935** may include instructions executable by the processor **940** to cause the device **905** to perform various aspects of signaling for inter-base station interference estimation as described herein, or the processor **940** and the memory **930** may be otherwise configured to perform or support such operations.

(139) FIG. **10** shows a block diagram **1000** of a device **1005** that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure. The device **1005** may be an example of aspects of a base station **105** as described herein. The device **1005** may include a receiver **1010**, a transmitter **1015**, and a communications manager **1020**. The device **1005** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).

(140) The receiver **1010** may provide a means for receiving information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to signaling for inter-base station interference estimation). Information may be passed on to other components of the device **1005**.

The receiver **1010** may utilize a single antenna or a set of multiple antennas.

(141) The transmitter **1015** may provide a means for transmitting signals generated by other components of the device **1005**. For example, the transmitter **1015** may transmit information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to signaling for inter-base station interference estimation). In some examples, the transmitter **1015** may be co-located with a receiver **1010** in a transceiver module. The transmitter **1015** may utilize a single antenna or a set of multiple antennas.

(142) The communications manager **1020**, the receiver **1010**, the transmitter **1015**, or various combinations thereof or various components thereof may be examples of means for performing various aspects of signaling for inter-base station interference estimation as described herein. For example, the communications manager **1020**, the receiver **1010**, the transmitter **1015**, or various combinations or components thereof may support a method for performing one or more of the functions described herein.

(143) In some examples, the communications manager **1020**, the receiver **1010**, the transmitter **1015**, or various combinations or components thereof may be implemented in hardware (e.g., in communications management circuitry). The hardware may include a processor, a DSP, an ASIC, an FPGA or other programmable logic device, a discrete gate or transistor logic, discrete hardware components, or any combination thereof configured as or otherwise supporting a means for performing the functions described in the present disclosure. In some examples, a processor and memory coupled with the processor may be configured to perform one or more of the functions described herein (e.g., by executing, by the processor, instructions stored in the memory).

(144) Additionally or alternatively, in some examples, the communications manager **1020**, the receiver **1010**, the transmitter **1015**, or various combinations or components thereof may be implemented in code (e.g., as communications management software or firmware) executed by a processor. If implemented in code executed by a processor, the functions of the communications manager **1020**, the receiver **1010**, the transmitter **1015**, or various combinations or components thereof may be performed by a general-purpose processor, a DSP, a CPU, an ASIC, an FPGA, or any combination of these or other programmable logic devices (e.g., configured as or otherwise supporting a means for performing the functions described in the present disclosure).

(145) In some examples, the communications manager **1020** may be configured to perform various operations (e.g., receiving, monitoring, transmitting) using or otherwise in cooperation with the receiver **1010**, the transmitter **1015**, or both. For example, the communications manager **1020** may receive information from the receiver **1010**, send information to the transmitter **1015**, or be integrated in combination with the receiver **1010**, the transmitter **1015**, or both to receive information, transmit information, or perform various other operations as described herein.

(146) The communications manager **1020** may support wireless communication at a first base station in accordance with examples as disclosed herein. For example, the communications manager **1020** may be configured as or otherwise support a means for transmitting, to a UE, a message for estimating a CLI channel between the first base station and a second base station, the first base station configured to receive uplink communications over the one or more resources. The communications manager **1020** may be configured as or otherwise support a means for monitoring the one or more resources for CLI associated with the second base station based on transmitting the message. The communications manager **1020** may be configured as or otherwise support a means for processing a first uplink communication from the UE communicated over a resource full duplexed with one or more downlink communications associated with the second base station based on monitoring the one or more resources for the CLI.

(147) Additionally, or alternatively, the communications manager **1020** may support wireless communication at a first base station in accordance with examples as disclosed herein. For example, the communications manager **1020** may be configured as or otherwise support a means

for transmitting a message indicating a first resource including a first type of resource or a second resource including a second type of resource to use for estimating a CLI channel between the first base station and a second base station, where the first base station is configured to receive uplink communications or transmit downlink communications to a UE over the first type of resource and the first base station is configured to transmit the downlink communications to the UE over the second type of resource. The communications manager **1020** may be configured as or otherwise support a means for monitoring one or more resources for CLI associated with the second base station based on transmitting the message. The communications manager **1020** may be configured as or otherwise support a means for processing a first uplink communication from the UE over a full-duplex resource based on monitoring the one or more resources for the CLI.

(148) By including or configuring the communications manager **1020** in accordance with examples as described herein, the device **1005** (e.g., a processor controlling or otherwise coupled to the receiver **1010**, the transmitter **1015**, the communications manager **1020**, or a combination thereof) may support techniques for more efficient utilization of communication resources and more accurate CLI channel estimation.

(149) FIG. **11** shows a block diagram **1100** of a device **1105** that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure. The device **1105** may be an example of aspects of a device **1005** or a base station **105** as described herein. The device **1105** may include a receiver **1110**, a transmitter **1115**, and a communications manager **1120**. The device **1105** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).

(150) The receiver **1110** may provide a means for receiving information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to signaling for inter-base station interference estimation). Information may be passed on to other components of the device **1105**. The receiver **1110** may utilize a single antenna or a set of multiple antennas.

(151) The transmitter **1115** may provide a means for transmitting signals generated by other components of the device **1105**. For example, the transmitter **1115** may transmit information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to signaling for inter-base station interference estimation). In some examples, the transmitter **1115** may be co-located with a receiver **1110** in a transceiver module. The transmitter **1115** may utilize a single antenna or a set of multiple antennas.

(152) The device **1105**, or various components thereof, may be an example of means for performing various aspects of signaling for inter-base station interference estimation as described herein. For example, the communications manager **1120** may include a messaging manager **1125** an interference manager **1130**, or any combination thereof. The communications manager **1120** may be an example of aspects of a communications manager **1020** as described herein. In some examples, the communications manager **1120**, or various components thereof, may be configured to perform various operations (e.g., receiving, monitoring, transmitting) using or otherwise in cooperation with the receiver **1110**, the transmitter **1115**, or both. For example, the communications manager **1120** may receive information from the receiver **1110**, send information to the transmitter **1115**, or be integrated in combination with the receiver **1110**, the transmitter **1115**, or both to receive information, transmit information, or perform various other operations as described herein.

(153) The communications manager **1120** may support wireless communication at a first base station in accordance with examples as disclosed herein. The messaging manager **1125** may be configured as or otherwise support a means for transmitting, to a UE, a message for estimating a CLI channel between the first base station and a second base station, the first base station configured to receive uplink communications over the one or more resources. The interference manager **1130** may be configured as or otherwise support a means for monitoring the one or more

resources for CLI associated with the second base station based on transmitting the message. The interference manager **1130** may be configured as or otherwise support a means for processing a first uplink communication from the UE communicated over a resource full duplexed with one or more downlink communications associated with the second base station based on monitoring the one or more resources for the CLI.

(154) Additionally, or alternatively, the communications manager **1120** may support wireless communication at a first base station in accordance with examples as disclosed herein. The messaging manager **1125** may be configured as or otherwise support a means for transmitting a message indicating a first resource including a first type of resource or a second resource including a second type of resource to use for estimating a CLI channel between the first base station and a second base station, where the first base station is configured to receive uplink communications or transmit downlink communications to a UE over the first type of resource and the first base station is configured to transmit the downlink communications to the UE over the second type of resource. The interference manager **1130** may be configured as or otherwise support a means for monitoring one or more resources for CLI associated with the second base station based on transmitting the message. The interference manager **1130** may be configured as or otherwise support a means for processing a first uplink communication from the UE over a full-duplex resource based on monitoring the one or more resources for the CLI.

(155) FIG. **12** shows a block diagram **1200** of a communications manager **1220** that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure. The communications manager **1220** may be an example of aspects of a communications manager **1020**, a communications manager **1120**, or both, as described herein. The communications manager **1220**, or various components thereof, may be an example of means for performing various aspects of signaling for inter-base station interference estimation as described herein. For example, the communications manager **1220** may include a messaging manager **1225**, an interference manager **1230**, a power manager **1235**, a frequency manager **1240**, a hopping pattern manager **1245**, a TA manager **1250**, a beam manager **1255**, or any combination thereof. Each of these components may communicate, directly or indirectly, with one another (e.g., via one or more buses).

(156) The communications manager **1220** may support wireless communication at a first base station in accordance with examples as disclosed herein. The messaging manager **1225** may be configured as or otherwise support a means for transmitting, to a UE, a message for estimating a CLI channel between the first base station and a second base station, the first base station configured to receive uplink communications over the one or more resources. The interference manager **1230** may be configured as or otherwise support a means for monitoring the one or more resources for CLI associated with the second base station based on transmitting the message. In some examples, the interference manager **1230** may be configured as or otherwise support a means for processing a first uplink communication from the UE communicated over a resource full duplexed with one or more downlink communications associated with the second base station based on monitoring the one or more resources for the CLI.

(157) In some examples, the messaging manager **1225** may be configured as or otherwise support a means for transmitting a second message to activate a communication pattern of the one or more communication patterns, where monitoring the one or more resources for the CLI is based on transmitting the second message.

(158) In some examples, the message includes a radio resource control message. In some examples, the second message includes a medium access control-control element or downlink control information.

(159) In some examples, the power manager **1235** may be configured as or otherwise support a means for identifying one or more UEs to mute the uplink communications during at least a portion of the one or more resources, the one or more UEs including the first UE, where transmitting the

message is based on identifying the one or more UEs.
(160) In some examples, the power manager **1235** may be configured as or otherwise support a means for identifying one or more UEs to reduce a transmit power of the uplink communications during at least a portion of the one or more resources, the one or more UEs including the first UE, where transmitting the message is based on identifying the one or more UEs.
(161) In some examples, the frequency manager **1240** may be configured as or otherwise support a means for identifying one or more subbands of frequency resources or one or more slots of time resources or both for estimating the CLI channel, where the message includes an indication of the one or more subbands or the one or more slots.
(162) In some examples, the one or more resources include one or more subbands of frequency resources and one or more slots of time resources. In some examples, the one or more communication patterns indicate to the UE to mute or reduce a transmit power of one or more uplink communications during a set of time resources of the one or more resources.
(163) In some examples, the frequency manager **1240** may be configured as or otherwise support a means for identifying a first subband of a BWP for estimating the CLI channel, a second subband of the BWP for an uplink communication by the UE, and a third subband of the BWP as a guard band between the first subband and the second subband, the one or more resources including the BWP, where the one or more communication patterns included in the message indicate the first subband, the second subband, and the third subband. In some examples, monitoring the one or more resources further includes monitoring the first subband of the BWP for the CLI.
(164) In some examples, the frequency manager **1240** may be configured as or otherwise support a means for receiving, over the second subband, the uplink communication from the UE based on transmitting the message.
(165) In some examples, the hopping pattern manager **1245** may be configured as or otherwise support a means for identifying a hopping pattern of subbands of a BWP for estimating the CLI channel and receiving the uplink communications, the hopping pattern spanning a set of multiple slots, where the one or more communication patterns indicate the hopping pattern.
(166) In some examples, the hopping pattern indicates at least one time resource for each subband of the subbands of the BWP for estimating the CLI channel.
(167) In some examples, the TA manager **1250** may be configured as or otherwise support a means for identifying a TA for an uplink communication communicated by the UE during the one or more resources, the TA based on a propagation delay of a downlink communication transmitted by the second base station. In some examples, the TA manager **1250** may be configured as or otherwise support a means for transmitting, to the UE, a second message that includes the TA, where the monitoring is based on transmitting the second message. In some examples, the TA is based on a capability of the UE.
(168) In some examples, the beam manager **1255** may be configured as or otherwise support a means for identifying one or more beams to restrict the UE from using to transmit the uplink communications during the one or more resources, where the message indicates the one or more beams.
(169) In some examples, the one or more communication patterns indicate to the UE a set of beams that the UE is restricted from using for the uplink communications.
(170) In some examples, the interference manager **1230** may be configured as or otherwise support a means for identifying a first quantity of reference signals transmitted by the second base station. In some examples, the interference manager **1230** may be configured as or otherwise support a means for identifying a second quantity of ports for estimating the CLI channel based on identifying the first quantity of reference signals, where the one or more communication patterns are based on identifying the second quantity of ports. In some examples, the second base station is configured to transmit downlink communications over the one or more resources.
(171) Additionally, or alternatively, the communications manager **1220** may support wireless

communication at a first base station in accordance with examples as disclosed herein. In some examples, the messaging manager **1225** may be configured as or otherwise support a means for transmitting a message indicating a first resource including a first type of resource or a second resource including a second type of resource to use for estimating a CLI channel between the first base station and a second base station, where the first base station is configured to receive uplink communications or transmit downlink communications to a UE over the first type of resource and the first base station is configured to transmit the downlink communications to the UE over the second type of resource. In some examples, the interference manager **1230** may be configured as or otherwise support a means for monitoring one or more resources for CLI associated with the second base station based on transmitting the message. In some examples, the interference manager **1230** may be configured as or otherwise support a means for processing a first uplink communication from the UE over a full-duplex resource based on monitoring the one or more resources for the CLI.

(172) In some examples, the messaging manager **1225** may be configured as or otherwise support a means for transmitting, to the UE, a second message to convert the first type of resource to the second type of resource, where monitoring the one or more resources for the CLI is based on transmitting the message.

(173) In some examples, the second message includes a slot format indicator configured to convert the first resource including the first type of resource to be the second type of resource.

(174) In some examples, the first type of resource includes a flexible symbol configured to use with the uplink communications or the downlink communications. In some examples, the second type of resource includes a downlink symbol.

(175) FIG. **13** shows a diagram of a system **1300** including a device **1305** that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure. The device **1305** may be an example of or include the components of a device **1005**, a device **1105**, or a base station **105** as described herein. The device **1305** may communicate wirelessly with one or more base stations **105**, UEs **115**, or any combination thereof. The device **1305** may include components for bi-directional voice and data communications including components for transmitting and receiving communications, such as a communications manager **1320**, a network communications manager **1310**, a transceiver **1315**, an antenna **1325**, a memory **1330**, code **1335**, a processor **1340**, and an inter-station communications manager **1345**. These components may be in electronic communication or otherwise coupled (e.g., operatively, communicatively, functionally, electronically, electrically) via one or more buses (e.g., a bus **1350**).

(176) The network communications manager **1310** may manage communications with a core network **130** (e.g., via one or more wired backhaul links). For example, the network communications manager **1310** may manage the transfer of data communications for client devices, such as one or more UEs **115**.

(177) In some cases, the device **1305** may include a single antenna **1325**. However, in some other cases the device **1305** may have more than one antenna **1325**, which may be capable of concurrently transmitting or receiving multiple wireless transmissions. The transceiver **1315** may communicate bi-directionally, via the one or more antennas **1325**, wired, or wireless links as described herein. For example, the transceiver **1315** may represent a wireless transceiver and may communicate bi-directionally with another wireless transceiver. The transceiver **1315** may also include a modem to modulate the packets, to provide the modulated packets to one or more antennas **1325** for transmission, and to demodulate packets received from the one or more antennas **1325**. The transceiver **1315**, or the transceiver **1315** and one or more antennas **1325**, may be an example of a transmitter **1015**, a transmitter **1115**, a receiver **1010**, a receiver **1110**, or any combination thereof or component thereof, as described herein.

(178) The memory **1330** may include RAM and ROM. The memory **1330** may store computer-readable, computer-executable code **1335** including instructions that, when executed by the

processor **1340**, cause the device **1305** to perform various functions described herein. The code **1335** may be stored in a non-transitory computer-readable medium such as system memory or another type of memory. In some cases, the code **1335** may not be directly executable by the processor **1340** but may cause a computer (e.g., when compiled and executed) to perform functions described herein. In some cases, the memory **1330** may contain, among other things, a BIOS which may control basic hardware or software operation such as the interaction with peripheral components or devices.

(179) The processor **1340** may include an intelligent hardware device (e.g., a general-purpose processor, a DSP, a CPU, a microcontroller, an ASIC, an FPGA, a programmable logic device, a discrete gate or transistor logic component, a discrete hardware component, or any combination thereof). In some cases, the processor **1340** may be configured to operate a memory array using a memory controller. In some other cases, a memory controller may be integrated into the processor **1340**. The processor **1340** may be configured to execute computer-readable instructions stored in a memory (e.g., the memory **1330**) to cause the device **1305** to perform various functions (e.g., functions or tasks supporting signaling for inter-base station interference estimation). For example, the device **1305** or a component of the device **1305** may include a processor **1340** and memory **1330** coupled to the processor **1340**, the processor **1340** and memory **1330** configured to perform various functions described herein.

(180) The inter-station communications manager **1345** may manage communications with other base stations **105**, and may include a controller or scheduler for controlling communications with UEs **115** in cooperation with other base stations **105**. For example, the inter-station communications manager **1345** may coordinate scheduling for transmissions to UEs **115** for various interference mitigation techniques such as beamforming or joint transmission. In some examples, the inter-station communications manager **1345** may provide an X2 interface within an LTE/LTE-A wireless communications network technology to provide communication between base stations **105**.

(181) The communications manager **1320** may support wireless communication at a first base station in accordance with examples as disclosed herein. For example, the communications manager **1320** may be configured as or otherwise support a means for transmitting, to a UE, a message for estimating a CLI channel between the first base station and a second base station, the first base station configured to receive uplink communications over the one or more resources. The communications manager **1320** may be configured as or otherwise support a means for monitoring the one or more resources for CLI associated with the second base station based on transmitting the message. The communications manager **1320** may be configured as or otherwise support a means for processing a first uplink communication from the UE communicated over a resource full duplexed with one or more downlink communications associated with the second base station based on monitoring the one or more resources for the CLI.

(182) Additionally, or alternatively, the communications manager **1320** may support wireless communication at a first base station in accordance with examples as disclosed herein. For example, the communications manager **1320** may be configured as or otherwise support a means for transmitting a message indicating a first resource including a first type of resource or a second resource including a second type of resource to use for estimating a CLI channel between the first base station and a second base station, where the first base station is configured to receive uplink communications or transmit downlink communications to a UE over the first type of resource and the first base station is configured to transmit the downlink communications to the UE over the second type of resource. The communications manager **1320** may be configured as or otherwise support a means for monitoring one or more resources for CLI associated with the second base station based on transmitting the message. The communications manager **1320** may be configured as or otherwise support a means for processing a first uplink communication from the UE over a full-duplex resource based on monitoring the one or more resources for the CLI.

(183) By including or configuring the communications manager **1320** in accordance with examples as described herein, the device **1305** may support techniques for more efficient utilization of communication resources and improved user experience related to reduced interference.
(184) In some examples, the communications manager **1320** may be configured to perform various operations (e.g., receiving, monitoring, transmitting) using or otherwise in cooperation with the transceiver **1315**, the one or more antennas **1325**, or any combination thereof. Although the communications manager **1320** is illustrated as a separate component, in some examples, one or more functions described with reference to the communications manager **1320** may be supported by or performed by the processor **1340**, the memory **1330**, the code **1335**, or any combination thereof. For example, the code **1335** may include instructions executable by the processor **1340** to cause the device **1305** to perform various aspects of signaling for inter-base station interference estimation as described herein, or the processor **1340** and the memory **1330** may be otherwise configured to perform or support such operations.
(185) FIG. **14** shows a flowchart illustrating a method **1400** that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure. The operations of the method **1400** may be implemented by a UE or its components as described herein. For example, the operations of the method **1400** may be performed by a UE **115** as described with reference to FIGS. **1** through **9**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the described functions. Additionally, or alternatively, the UE may perform aspects of the described functions using special-purpose hardware.
(186) At **1405**, the method may include receiving, from a first base station, a message that indicates one or more communication patterns of one or more resources for estimating a CLI channel between uplink communications associated with the first base station and downlink communications associated with a second base station. The operations of **1405** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1405** may be performed by a messaging manager **825** as described with reference to FIG. **8**.
(187) At **1410**, the method may include adjusting a transmission parameter for an uplink communication based on receiving the message from the first base station. The operations of **1410** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1410** may be performed by an interference manager **830** as described with reference to FIG. **8**.
(188) At **1415**, the method may include transmitting the uplink communication over the one or more resources based on adjusting the transmission parameter. The operations of **1415** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1415** may be performed by an uplink manager **835** as described with reference to FIG. **8**.
(189) FIG. **15** shows a flowchart illustrating a method **1500** that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure. The operations of the method **1500** may be implemented by a UE or its components as described herein. For example, the operations of the method **1500** may be performed by a UE **115** as described with reference to FIGS. **1** through **9**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the described functions. Additionally, or alternatively, the UE may perform aspects of the described functions using special-purpose hardware.
(190) At **1505**, the method may include receiving, from a first base station, a message that indicates one or more communication patterns of one or more resources for estimating a CLI channel between uplink communications associated with the first base station and downlink communications associated with a second base station. The operations of **1505** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of

**1505** may be performed by a messaging manager **825** as described with reference to FIG. **8**.
(191) At **1510**, the method may include receiving, from the first base station, a second message to activate a communication pattern of the one or more communication patterns. The operations of **1510** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1510** may be performed by a messaging manager **825** as described with reference to FIG. **8**.
(192) At **1515**, the method may include adjusting a transmission parameter for an uplink communication based on receiving the message from the first base station and based on receiving the second message from the first base station. The operations of **1515** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1515** may be performed by an interference manager **830** as described with reference to FIG. **8**.
(193) At **1520**, the method may include transmitting the uplink communication over the one or more resources based on adjusting the transmission parameter. The operations of **1520** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1520** may be performed by an uplink manager **835** as described with reference to FIG. **8**.
(194) FIG. **16** shows a flowchart illustrating a method **1600** that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure. The operations of the method **1600** may be implemented by a base station or its components as described herein. For example, the operations of the method **1600** may be performed by a base station **105** as described with reference to FIGS. **1** through **5** and **10** through **13**. In some examples, a base station may execute a set of instructions to control the functional elements of the base station to perform the described functions. Additionally, or alternatively, the base station may perform aspects of the described functions using special-purpose hardware.
(195) At **1605**, the method may include transmitting, to a UE, a message that indicates one or more communication patterns of one or more resources for estimating a CLI channel between the first base station and a second base station, the first base station configured to receive uplink communications over the one or more resources. The operations of **1605** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1605** may be performed by a messaging manager **1225** as described with reference to FIG. **12**.
(196) At **1610**, the method may include monitoring the one or more resources for CLI associated with the second base station based on transmitting the message. The operations of **1610** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1610** may be performed by an interference manager **1230** as described with reference to FIG. **12**.
(197) At **1615**, the method may include processing a first uplink communication from the UE communicated over a resource full duplexed with one or more downlink communications associated with the second base station based on monitoring the one or more resources for the CLI. The operations of **1615** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1615** may be performed by an interference manager **1230** as described with reference to FIG. **12**.
(198) FIG. **17** shows a flowchart illustrating a method **1700** that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure. The operations of the method **1700** may be implemented by a base station or its components as described herein. For example, the operations of the method **1700** may be performed by a base station **105** as described with reference to FIGS. **1** through **5** and **10** through **13**. In some examples, a base station may execute a set of instructions to control the functional elements of the base station to perform the described functions. Additionally, or alternatively, the base station may perform aspects of the described functions using special-purpose hardware.
(199) At **1705**, the method may include transmitting, to a UE, a message that indicates one or more

communication patterns of one or more resources for estimating a CLI channel between the first base station and a second base station, the first base station configured to receive uplink communications over the one or more resources. The operations of **1705** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1705** may be performed by a messaging manager **1225** as described with reference to FIG. **12**.

(200) At **1710**, the method may include transmitting a second message to activate a communication pattern of the one or more communication patterns. The operations of **1710** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1710** may be performed by a messaging manager **1225** as described with reference to FIG. **12**.

(201) At **1715**, the method may include monitoring the one or more resources for CLI associated with the second base station based on transmitting the message and transmitting the second message. The operations of **1715** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1715** may be performed by an interference manager **1230** as described with reference to FIG. **12**.

(202) At **1720**, the method may include processing a first uplink communication from the UE communicated over a resource full duplexed with one or more downlink communications associated with the second base station based on monitoring the one or more resources for the CLI. The operations of **1720** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1720** may be performed by an interference manager **1230** as described with reference to FIG. **12**.

(203) FIG. **18** shows a flowchart illustrating a method **1800** that supports signaling for inter-base station interference estimation in accordance with aspects of the present disclosure. The operations of the method **1800** may be implemented by a base station or its components as described herein. For example, the operations of the method **1800** may be performed by a base station **105** as described with reference to FIGS. **1** through **5** and **10** through **13**. In some examples, a base station may execute a set of instructions to control the functional elements of the base station to perform the described functions. Additionally, or alternatively, the base station may perform aspects of the described functions using special-purpose hardware.

(204) At **1805**, the method may include transmitting a message indicating a first resource including a first type of resource or a second resource including a second type of resource to use for estimating a CLI channel between the first base station and a second base station, where the first base station is configured to receive uplink communications or transmit downlink communications to a UE over the first type of resource and the first base station is configured to transmit the downlink communications to the UE over the second type of resource. The operations of **1805** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1805** may be performed by a messaging manager **1225** as described with reference to FIG. **12**.

(205) At **1810**, the method may include monitoring one or more resources for CLI associated with the second base station based on transmitting the message. The operations of **1810** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1810** may be performed by an interference manager **1230** as described with reference to FIG. **12**.

(206) At **1815**, the method may include processing a first uplink communication from the UE over a full-duplex resource based on monitoring the one or more resources for the CLI. The operations of **1815** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1815** may be performed by an interference manager **1230** as described with reference to FIG. **12**.

(207) The following provides an overview of aspects of the present disclosure:

(208) Aspect 1: A method for wireless communication at a UE, comprising: receiving, from a first base station, a message that indicates one or more communication patterns of one or more

resources for estimating a CLI channel between uplink communications associated with the first base station and downlink communications associated with a second base station; adjusting a transmission parameter for an uplink communication based at least in part on receiving the message from the first base station; and transmitting the uplink communication over the one or more resources based at least in part on adjusting the transmission parameter.

(209) Aspect 2: The method of aspect 1, further comprising: receiving, from the first base station, a second message to activate a communication pattern of the one or more communication patterns, wherein adjusting a transmission parameter for the uplink communication is based at least in part on receiving the second message.

(210) Aspect 3: The method of aspect 2, wherein the message comprises a radio resource control message; and the second message comprises a medium access control-control element or downlink control information.

(211) Aspect 4: The method of any of aspects 1 through 3, wherein adjusting the transmission parameter for the uplink communication further comprises: reducing the transmit power of the uplink communications during at least a portion of the one or more resources.

(212) Aspect 5: The method of aspect 4, wherein reducing the transmit power of the uplink communications further comprises: reducing the transmit power of the uplink communications to zero during the portion of the one or more resources.

(213) Aspect 6: The method of any of aspects 1 through 5, further comprising: identifying a first subband of a BWP for estimating the CLI channel, a second subband of the BWP for a second uplink communication by the UE, and a third subband of the BWP as a guard band between the first subband and the second subband based at least in part on receiving the message, the one or more resources comprising the BWP, wherein the one or more communication patterns included in the message indicate the first subband, the second subband, and the third subband, wherein adjusting a transmission parameter for the uplink communication is based at least in part on the identifying.

(214) Aspect 7: The method of aspect 6, further comprising: transmitting, over the second subband, the uplink communication to the first base station based at least in part on transmitting the message.

(215) Aspect 8: The method of any of aspects 1 through 7, further comprising: identifying a hopping pattern of subbands of a BWP for estimating the CLI channel and transmitting the uplink communications based at least in part on receiving the message, the hopping pattern spanning a plurality of slots, wherein the one or more communication patterns indicate the hopping pattern, wherein adjusting a transmission parameter for the uplink communication is based at least in part on the identifying.

(216) Aspect 9: The method of aspect 8, wherein the hopping pattern indicates at least one time resource for each subband of the subbands of the BWP for estimating the CLI channel.

(217) Aspect 10: The method of any of aspects 1 through 9, wherein adjusting the transmission parameter for the uplink communication further comprises: applying a TA for the uplink communication transmitted by the UE during a first time resource of the one or more resources based at least in part on receiving the message, the TA based at least in part on a propagation delay of a downlink communication transmitted by the second base station.

(218) Aspect 11: The method of aspect 10, wherein the TA is based at least in part on a capability of the UE.

(219) Aspect 12: The method of any of aspects 1 through 11, wherein adjusting the transmission parameter for the uplink communication further comprises: identifying one or more beams to restrict the UE from using to transmit the uplink communication during the one or more resources, wherein the message indicates the one or more beams.

(220) Aspect 13: The method of aspect 12, further comprising: receiving, from the first base station, a second message that schedules the UE with the uplink communication using the one or more beams; and determining to skip the uplink communication or use a different beam than the

one or more beams based at least in part on the one or more communication patterns.
(221) Aspect 14: The method of any of aspects 1 through 13, further comprising: identifying one or more subbands of frequency resources or one or more slots of time resources or both for estimating the CLI channel, wherein the message includes an indication of the one or more subbands or the one or more slots.
(222) Aspect 15: The method of any of aspects 1 through 14, wherein the one or more resources comprise one or more subbands of frequency resources and one or more slots of time resources.
(223) Aspect 16: The method of any of aspects 1 through 15, wherein the one or more communication patterns indicate to the UE to mute or reduce a transmit power of one or more uplink communications during a set of time resources of the one or more resources.
(224) Aspect 17: The method of any of aspects 1 through 16, wherein the one or more communication patterns indicate a set of beams that the UE is restricted from using for the uplink communications.
(225) Aspect 18: The method of any of aspects 1 through 17, wherein the one or more communication patterns indicate one or more ports for estimating the CLI channel.
(226) Aspect 19: The method of any of aspects 1 through 18, wherein the first base station is configured to receive the uplink communications over the one or more resources; the second base station is configured to transmit the downlink communications over the one or more resources.
(227) Aspect 20: A method for wireless communication at a first base station, comprising: transmitting, to a UE, a message that indicates one or more communication patterns of one or more resources for estimating a CLI channel between the first base station and a second base station, the first base station configured to receive uplink communications over the one or more resources; monitoring the one or more resources for CLI associated with the second base station based at least in part on transmitting the message; and processing a first uplink communication from the UE communicated over a resource full duplexed with one or more downlink communications associated with the second base station based at least in part on monitoring the one or more resources for the CLI.
(228) Aspect 21: The method of aspect 20, further comprising: transmitting a second message to activate a communication pattern of the one or more communication patterns, wherein monitoring the one or more resources for the CLI is based at least in part on transmitting the second message.
(229) Aspect 22: The method of aspect 21, wherein the message comprises a radio resource control message; and the second message comprises a medium access control-control element or downlink control information.
(230) Aspect 23: The method of any of aspects 20 through 22, further comprising: identifying one or more UEs to mute the uplink communications during at least a portion of the one or more resources, the one or more UEs comprising the first UE, wherein transmitting the message is based at least in part on identifying the one or more UEs.
(231) Aspect 24: The method of any of aspects 20 through 23, further comprising: identifying one or more UEs to reduce a transmit power of the uplink communications during at least a portion of the one or more resources, the one or more UEs comprising the first UE, wherein transmitting the message is based at least in part on identifying the one or more UEs.
(232) Aspect 25: The method of any of aspects 20 through 24, further comprising: identifying one or more subbands of frequency resources or one or more slots of time resources or both for estimating the CLI channel, wherein the message includes an indication of the one or more subbands or the one or more slots.
(233) Aspect 26: The method of any of aspects 20 through 25, wherein the one or more resources comprise one or more subbands of frequency resources and one or more slots of time resources.
(234) Aspect 27: The method of any of aspects 20 through 26, wherein the one or more communication patterns indicate to the UE to mute or reduce a transmit power of one or more uplink communications during a set of time resources of the one or more resources.

(235) Aspect 28: The method of any of aspects 20 through 27, further comprising: identifying a first subband of a BWP for estimating the CLI channel, a second subband of the BWP for an uplink communication by the UE, and a third subband of the BWP as a guard band between the first subband and the second subband, the one or more resources comprising the BWP, wherein the one or more communication patterns included in the message indicate the first subband, the second subband, and the third subband.
(236) Aspect 29: The method of aspect 28, wherein monitoring the one or more resources further comprises monitoring the first subband of the BWP for the CLI.
(237) Aspect 30: The method of any of aspects 28 through 29, further comprising: receiving, over the second subband, the uplink communication from the UE based at least in part on transmitting the message.
(238) Aspect 31: The method of any of aspects 20 through 30, further comprising: identifying a hopping pattern of subbands of a BWP for estimating the CLI channel and receiving the uplink communications, the hopping pattern spanning a plurality of slots, wherein the one or more communication patterns indicate the hopping pattern.
(239) Aspect 32: The method of aspect 31, wherein the hopping pattern indicates at least one time resource for each subband of the subbands of the BWP for estimating the CLI channel.
(240) Aspect 33: The method of any of aspects 20 through 32, further comprising: identifying a TA for an uplink communication communicated by the UE during the one or more resources, the TA based at least in part on a propagation delay of a downlink communication transmitted by the second base station; and transmitting, to the UE, a second message that comprises the TA, wherein the monitoring is based at least in part on transmitting the second message.
(241) Aspect 34: The method of aspect 33, wherein the TA is based at least in part on a capability of the UE.
(242) Aspect 35: The method of any of aspects 20 through 34, further comprising: identifying one or more beams to restrict the UE from using to transmit the uplink communications during the one or more resources, wherein the message indicates the one or more beams.
(243) Aspect 36: The method of any of aspects 20 through 35, wherein the one or more communication patterns indicate to the UE a set of beams that the UE is restricted from using for the uplink communications.
(244) Aspect 37: The method of any of aspects 20 through 36, further comprising: identifying a first quantity of reference signals transmitted by the second base station; and identifying a second quantity of ports for estimating the CLI channel based at least in part on identifying the first quantity of reference signals, wherein the one or more communication patterns are based at least in part on identifying the second quantity of ports.
(245) Aspect 38: The method of any of aspects 20 through 37, wherein the second base station is configured to transmit downlink communications over the one or more resources.
(246) Aspect 39: A method for wireless communication at a first base station, comprising: transmitting a message indicating a first resource comprising a first type of resource or a second resource comprising a second type of resource to use for estimating a CLI channel between the first base station and a second base station, wherein the first base station is configured to receive uplink communications or transmit downlink communications to a UE over the first type of resource and the first base station is configured to transmit the downlink communications to the UE over the second type of resource; monitoring one or more resources for CLI associated with the second base station based at least in part on transmitting the message; and processing a first uplink communication from the UE over a full-duplex resource based at least in part on monitoring the one or more resources for the CLI.
(247) Aspect 40: The method of aspect 39, further comprising: transmitting, to the UE, a second message to convert the first type of resource to the second type of resource, wherein monitoring the one or more resources for the CLI is based at least in part on transmitting the message.

(248) Aspect 41: The method of aspect 40, wherein the second message comprises a slot format indicator configured to convert the first resource comprising the first type of resource to be the second type of resource.
(249) Aspect 42: The method of any of aspects 39 through 41, wherein the first type of resource comprises a flexible symbol configured to use with the uplink communications or the downlink communications; and the second type of resource comprises a downlink symbol.
(250) Aspect 43: An apparatus for wireless communication at a UE, comprising a memory; and a processor coupled to the memory and configured to cause the apparatus to perform a method of any of aspects 1 through 19.
(251) Aspect 44: An apparatus for wireless communication at a UE, comprising at least one means for performing a method of any of aspects 1 through 19.
(252) Aspect 45: A non-transitory computer-readable medium storing code for wireless communication at a UE, the code comprising instructions executable by a processor to perform a method of any of aspects 1 through 19.
(253) Aspect 46: An apparatus for wireless communication at a first base station, comprising a memory; and a processor coupled to the memory and configured to cause the apparatus to perform a method of any of aspects 20 through 38.
(254) Aspect 47: An apparatus for wireless communication at a first base station, comprising at least one means for performing a method of any of aspects 20 through 38.
(255) Aspect 48: A non-transitory computer-readable medium storing code for wireless communication at a first base station, the code comprising instructions executable by a processor to perform a method of any of aspects 20 through 38.
(256) Aspect 49: An apparatus for wireless communication at a first base station, comprising a memory; and a processor coupled to the memory and configured to cause the apparatus to cause the apparatus to perform a method of any of aspects 39 through 42.
(257) Aspect 50: An apparatus for wireless communication at a first base station, comprising at least one means for performing a method of any of aspects 39 through 42.
(258) Aspect 51: A non-transitory computer-readable medium storing code for wireless communication at a first base station, the code comprising instructions executable by a processor to perform a method of any of aspects 39 through 42.
(259) It should be noted that the methods described herein describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Further, aspects from two or more of the methods may be combined.
(260) Although aspects of an LTE, LTE-A, LTE-A Pro, or NR system may be described for purposes of example, and LTE, LTE-A, LTE-A Pro, or NR terminology may be used in much of the description, the techniques described herein are applicable beyond LTE, LTE-A, LTE-A Pro, or NR networks. For example, the described techniques may be applicable to various other wireless communications systems such as Ultra Mobile Broadband (UMB), Institute of Electrical and Electronics Engineers (IEEE) 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, Flash-OFDM, as well as other systems and radio technologies not explicitly mentioned herein.
(261) Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.
(262) The various illustrative blocks and components described in connection with the disclosure herein may be implemented or performed with a general-purpose processor, a DSP, an ASIC, a CPU, an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor

may be any processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

(263) The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described herein may be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.

(264) Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A non-transitory storage medium may be any available medium that may be accessed by a general-purpose or special-purpose computer. By way of example, and not limitation, non-transitory computer-readable media may include RAM, ROM, electrically erasable programmable ROM (EEPROM), flash memory, compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that may be used to carry or store desired program code means in the form of instructions or data structures and that may be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of computer-readable medium. Disk and disc, as used herein, include CD, laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above are also included within the scope of computer-readable media.

(265) As used herein, including in the claims, “or” as used in a list of items (e.g., a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an example step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

(266) The term “determine” or “determining” encompasses a wide variety of actions and, therefore, “determining” can include calculating, computing, processing, deriving, investigating, looking up (such as via looking up in a table, a database or another data structure), ascertaining and the like. Also, “determining” can include receiving (such as receiving information), accessing (such as accessing data in a memory) and the like. Also, “determining” can include resolving, selecting, choosing, establishing and other such similar actions.

(267) In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label, or other

subsequent reference label.
(268) The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term "example" used herein means "serving as an example, instance, or illustration," and not "preferred" or "advantageous over other examples." The detailed description includes specific details for the purpose of providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, known structures and devices are shown in block diagram form in order to avoid obscuring the concepts of the described examples.
(269) The description herein is provided to enable a person having ordinary skill in the art to make or use the disclosure. Various modifications to the disclosure will be apparent to a person having ordinary skill in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

## Claims

1. A user equipment (UE) for wireless communication, comprising: one or more processors; one or more memories coupled with the one or more processors; and instructions stored in the one or more memories and executable by the one or more processors to cause the UE to: receive, from a first base station, a message for estimating a cross-link interference channel between uplink communications associated with the first base station and downlink communications associated with a second base station; adjust a transmission parameter for an uplink communication based at least in part on receiving the message from the first base station; and transmit the uplink communication over one or more resources based at least in part on adjusting the transmission parameter.
2. The UE of claim 1, wherein the instructions are further executable by the one or more processors to cause the UE to: receive, from the first base station, a second message to activate a communication pattern of one or more communication patterns, wherein adjusting the transmission parameter for the uplink communication is based at least in part on receiving the second message.
3. The UE of claim 2, wherein: the message comprises a radio resource control message; and the second message comprises a medium access control-control element or downlink control information.
4. The UE of claim 1, wherein the instructions to adjust the transmission parameter for the uplink communication are further executable by the one or more processors to cause the UE to: reduce a transmit power of the uplink communications during at least a portion of the one or more resources.
5. The UE of claim 4, wherein the instructions to reduce the transmit power of the uplink communications are further executable by the one or more processors to cause the UE to: reduce the transmit power of the uplink communications to zero during the portion of the one or more resources.
6. The UE of claim 1, wherein the instructions are further executable by the one or more processors to cause the UE to: identify a first subband of a bandwidth part for estimating the cross-link interference channel, a second subband of the bandwidth part for a second uplink communication by the UE, and a third subband of the bandwidth part as a guard band between the first subband and the second subband based at least in part on receiving the message, the one or more resources comprising the bandwidth part, wherein adjusting the transmission parameter for the uplink communication is based at least in part on the identifying.
7. The UE of claim 6, wherein the instructions are further executable by the one or more processors to cause the UE to: transmit, over the second subband, the uplink communication to the first base

station based at least in part on receiving the message.
8. The UE of claim 1, wherein the instructions are further executable by the one or more processors to cause the UE to: identify a hopping pattern of subbands of a bandwidth part for estimating the cross-link interference channel and transmitting the uplink communications based at least in part on receiving the message, the hopping pattern spanning a plurality of slots, wherein adjusting the transmission parameter for the uplink communication is based at least in part on the identifying.
9. The UE of claim 1, wherein the instructions to adjust the transmission parameter for the uplink communication are further executable by the one or more processors to cause the UE to: apply a timing advance for the uplink communication transmitted by the UE during a first time resource of the one or more resources based at least in part on receiving the message, the timing advance based at least in part on a propagation delay of a downlink communication transmitted by the second base station.
10. The UE of claim 1, wherein the instructions to adjust the transmission parameter for the uplink communication are further executable by the one or more processors to cause the UE to: identify one or more beams to restrict the UE from using to transmit the uplink communication during the one or more resources, wherein the message indicates the one or more beams.
11. The UE of claim 10, wherein the instructions are further executable by the one or more processors to cause the UE to: receive, from the first base station, a second message that schedules the UE with the uplink communication using the one or more beams; and determine to skip the uplink communication or use a different beam than the one or more beams based at least in part on the message.
12. The UE of claim 1, wherein the instructions are further executable by the one or more processors to cause the UE to: identify one or more subbands of frequency resources or one or more slots of time resources or both for estimating the cross-link interference channel, wherein the message includes an indication of the one or more subbands or the one or more slots.
13. A first base station for wireless communication, comprising: one or more processors; one or more memories coupled with the one or more processors; and instructions stored in the one or more memories and executable by the one or more processors to cause the first base station to: transmit, to a user equipment (UE), a message for estimating a cross-link interference channel between the first base station and a second base station, the first base station configured to receive uplink communications over one or more resources; monitor the one or more resources for cross-link interference associated with the second base station based at least in part on transmitting the message; and process an uplink communication from the UE communicated over a resource full duplexed with one or more downlink communications associated with the second base station based at least in part on monitoring the one or more resources for the cross-link interference.
14. The first base station of claim 13, wherein the instructions are further executable by the one or more processors to cause the first base station to: transmit a second message to activate a communication pattern of one or more communication patterns, wherein monitoring the one or more resources for the cross-link interference is based at least in part on transmitting the second message, wherein the message comprises a radio resource control message and the second message comprises a medium access control-control element or downlink control information.
15. The first base station of claim 13, wherein the instructions are further executable by the one or more processors to cause the first base station to: identify one or more UEs to mute the uplink communications during at least a portion of one or more resources, the one or more UEs comprising the UE, wherein transmitting the message is based at least in part on identifying the one or more UEs.
16. The first base station of claim 13, wherein the instructions are further executable by the one or more processors to cause the first base station to: identify one or more subbands of frequency resources or one or more slots of time resources or both for estimating the cross-link interference channel, wherein the message includes an indication of the one or more subbands or the one or

more slots.

17. The first base station of claim 13, wherein the instructions are further executable by the one or more processors to cause the first base station to: identify a first subband of a bandwidth part for estimating the cross-link interference channel, a second subband of the bandwidth part for the uplink communication by the UE, and a third subband of the bandwidth part as a guard band between the first subband and the second subband, the one or more resources comprising the bandwidth part.

18. The first base station of claim 17, wherein the instructions to monitor the one or more resources are further executable by the one or more processors to cause the first base station to monitor the first subband of the bandwidth part for the cross-link interference.

19. The first base station of claim 17, wherein the instructions are further executable by the one or more processors to cause the first base station to: receive, over the second subband, the uplink communication from the UE based at least in part on transmitting the message.

20. A first base station for wireless communication, comprising: one or more processors; one or more memories coupled with the one or more processors; and instructions stored in the one or more memories and executable by the one or more processors to cause the first base station to: transmit a message indicating a first resource comprising a first type of resource or a second resource comprising a second type of resource to use for estimating a cross-link interference channel between the first base station and a second base station, wherein the first base station is configured to receive uplink communications or transmit downlink communications to a user equipment (UE) over the first type of resource and the first base station is configured to transmit the downlink communications to the UE over the second type of resource; monitor one or more resources for cross-link interference associated with the second base station based at least in part on transmitting the message; and process an uplink communication from the UE over a full-duplex resource based at least in part on monitoring the one or more resources for the cross-link interference.